Designing Colormaps

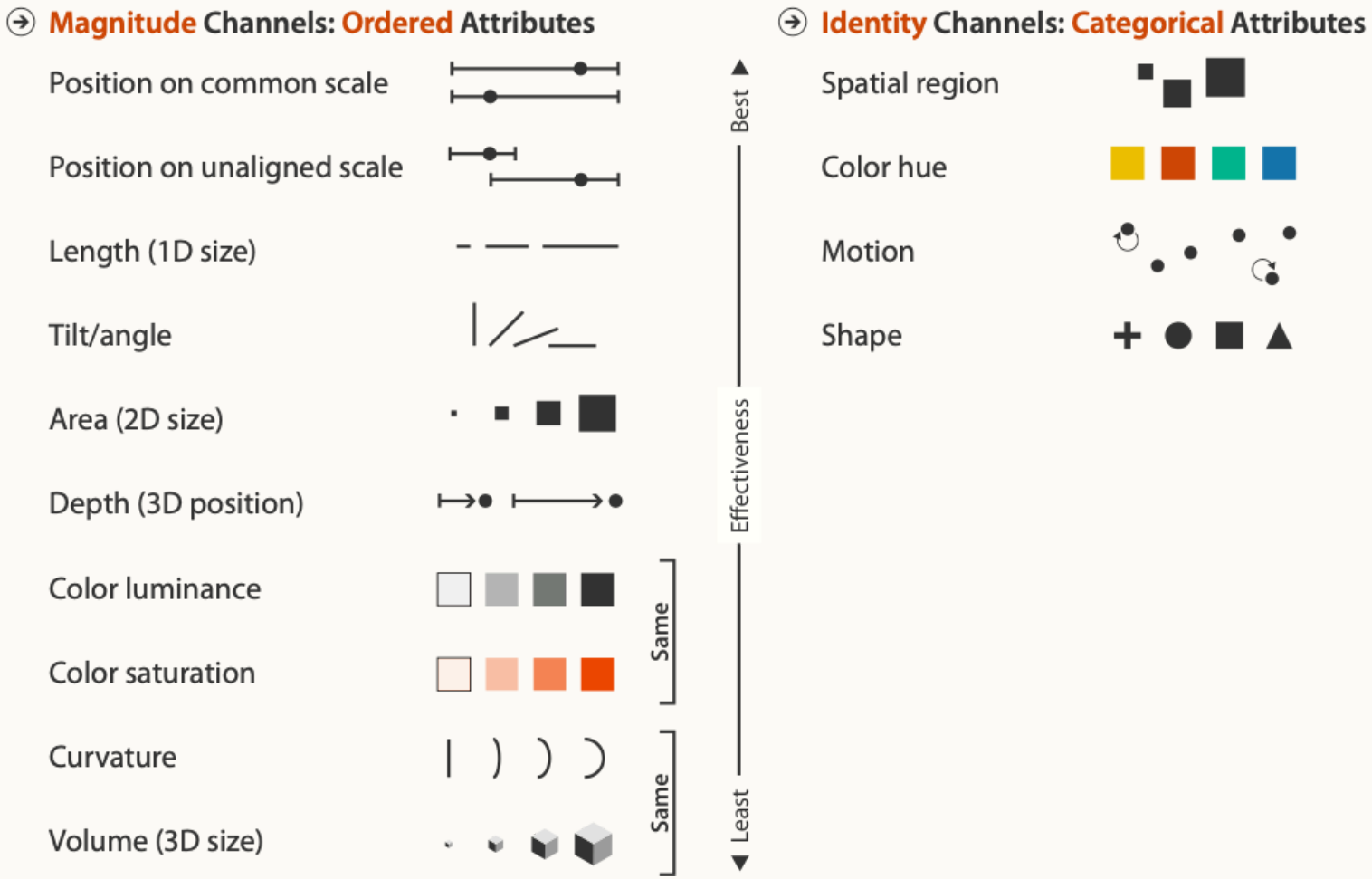
From data relationships to readable color

Information Visualization · Lecture 11 · Week 6

Agenda

01	**Color perception**	Vision, context, culture, comparison, and color models
02	**Colormap types**	Categorical, sequential, and diverging palettes
03	**Value mapping**	Domains, normalization, class boundaries, and legends
04	**Number of attributes**	Univariate and bivariate colormaps
05	**Accessible color**	Color-vision differences and redundant cues

Color is a multifunctional channel



Magnitude

Variations in lightness can show order and reveal patterns.

Identity

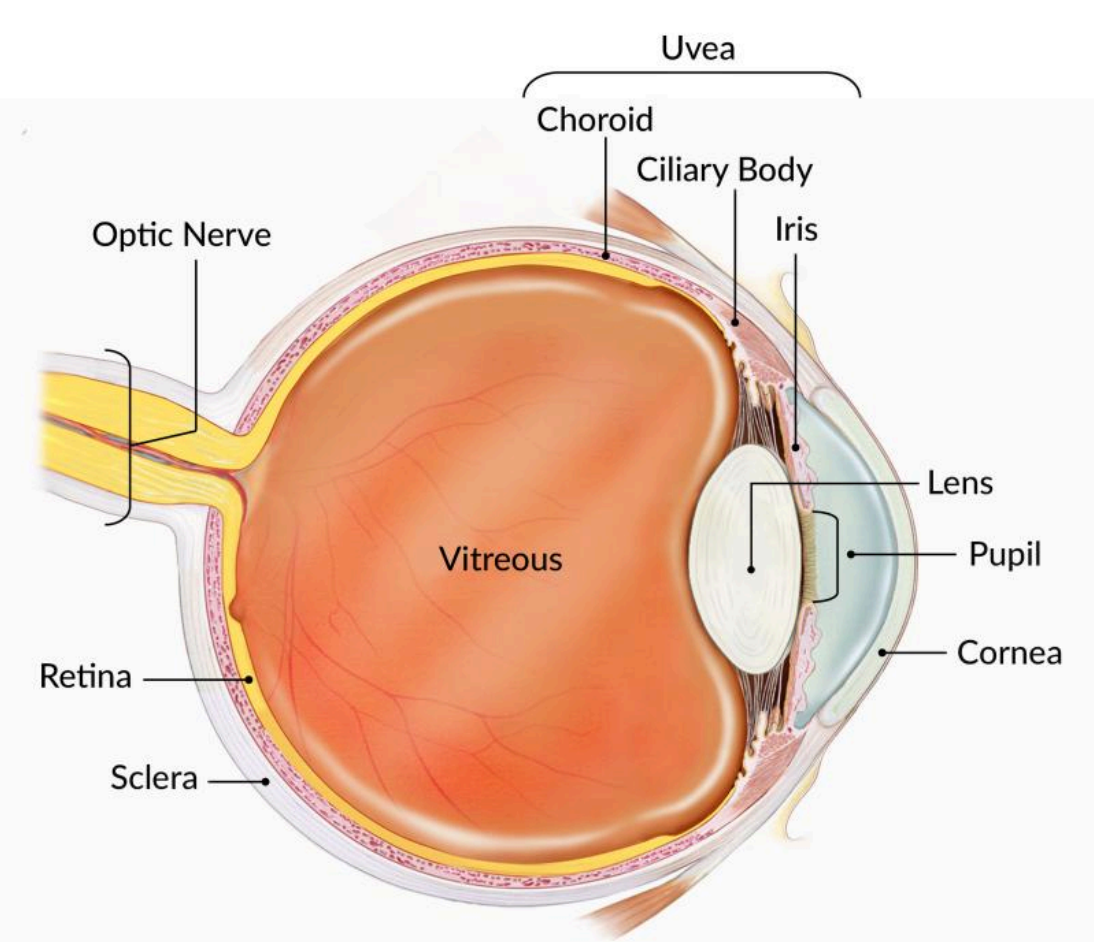
Differences in hue can distinguish categories.

Emphasis

Contrast can direct attention to selected items.

Tamara Munzner · Visualization Analysis and Design · UBC teaching materials

The eye focuses light onto the retina



The **retina** is the light-sensitive tissue lining the back of the eye.

The cornea and lens focus incoming light onto it. Retinal photoreceptors convert light into electrical signals.

Retinal circuits process these signals, which travel through the **optic nerve** to the brain.

Where the optic nerve leaves the retina, there are no photoreceptors.

National Eye Institute · How the Eyes Work · Exploratorium · Blind Spot

The blind spot reveals a gap in retinal sampling

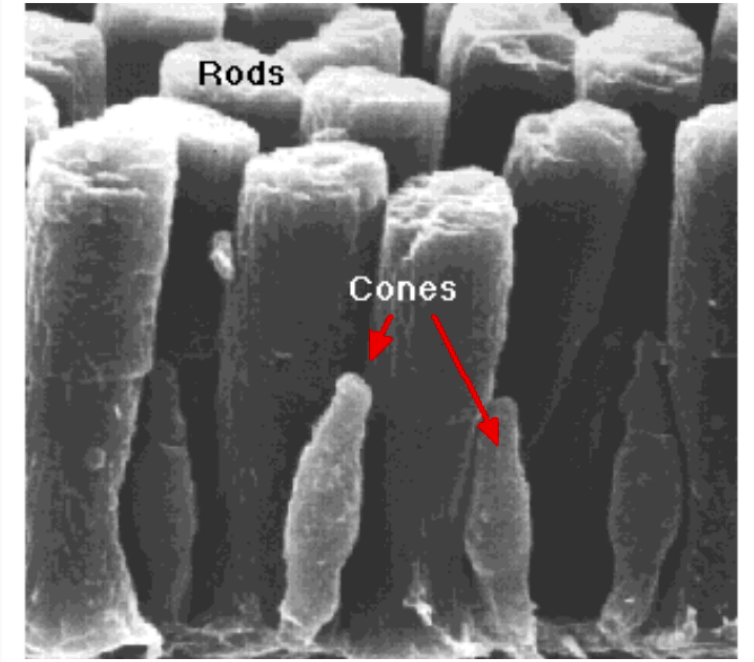
The **blind spot** is the region of the visual field corresponding to the optic disc, where the optic nerve exits the retina and no photoreceptors are present.

1. On your own screen or handout, close your **left eye** and keep your right eye fixed on **R**.
2. Start about three times the R–L gap away. Move closer or farther while keeping your gaze on R.
3. At a particular distance, **L disappears** even though the letter is still there.

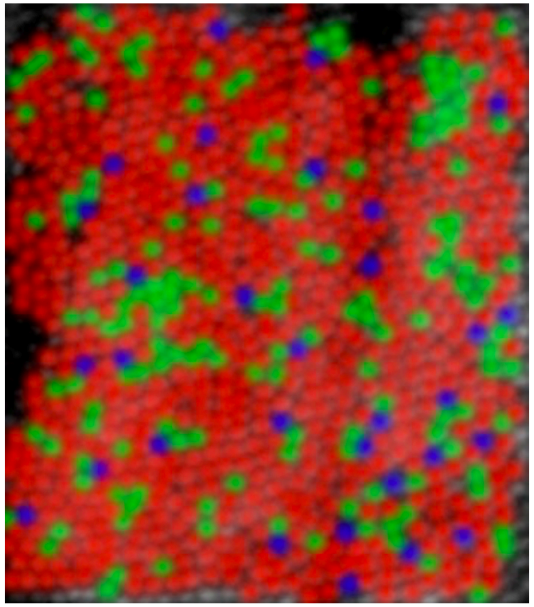
Exploratorium · Blind Spot

Rods and cones respond to different light levels

Photoreceptors are retinal cells that convert absorbed light into electrical signals. The two main types are rods and cones.



Rods: sensitive to dim light.
Cones: color and detail in brighter light.

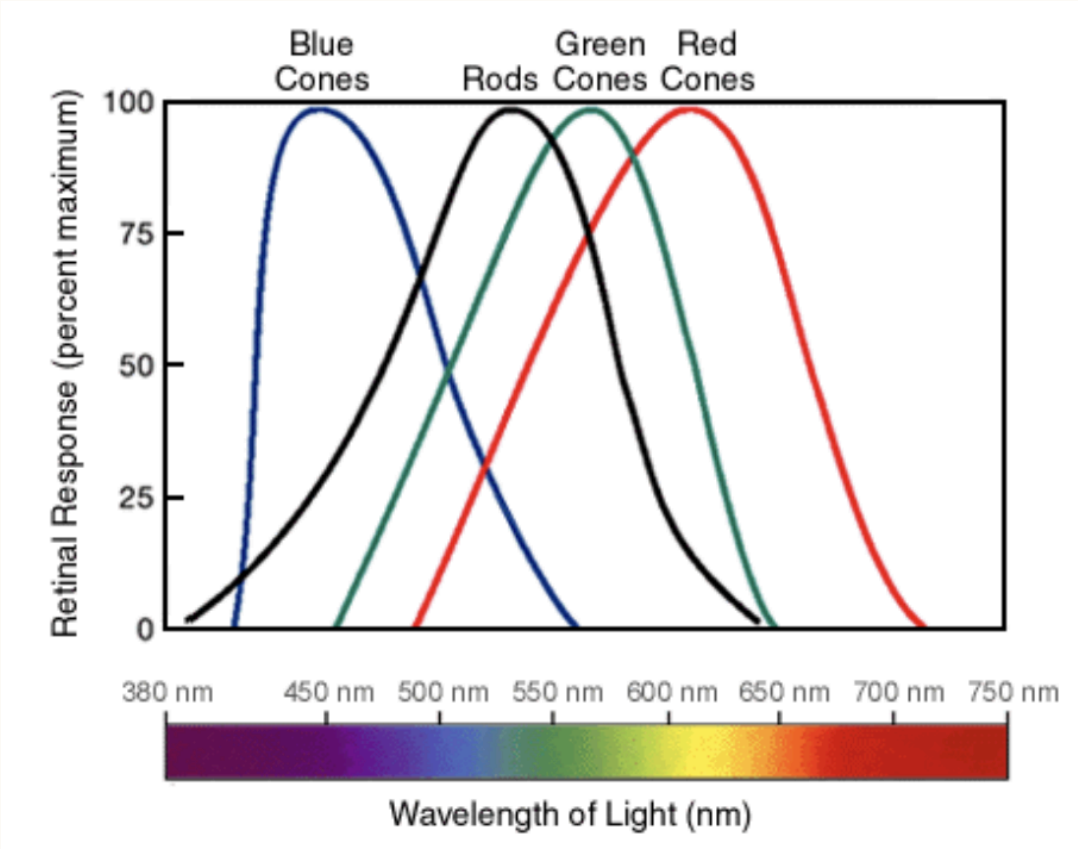


Three cone types: S, M, and L.
Colors indicate cone types: S (blue), M (green), L (red).

Reading: Helga Kolb · Webvision · Photoreceptors · Roorda & Williams · Nature (1999)

Cone sensitivities overlap across wavelengths

Spectral sensitivity describes a photoreceptor's relative response to equal amounts of light energy at different wavelengths.



Schematic response curves · Reading: Andrew Stockman · Cone fundamentals

One wavelength can stimulate **more than one cone type**.

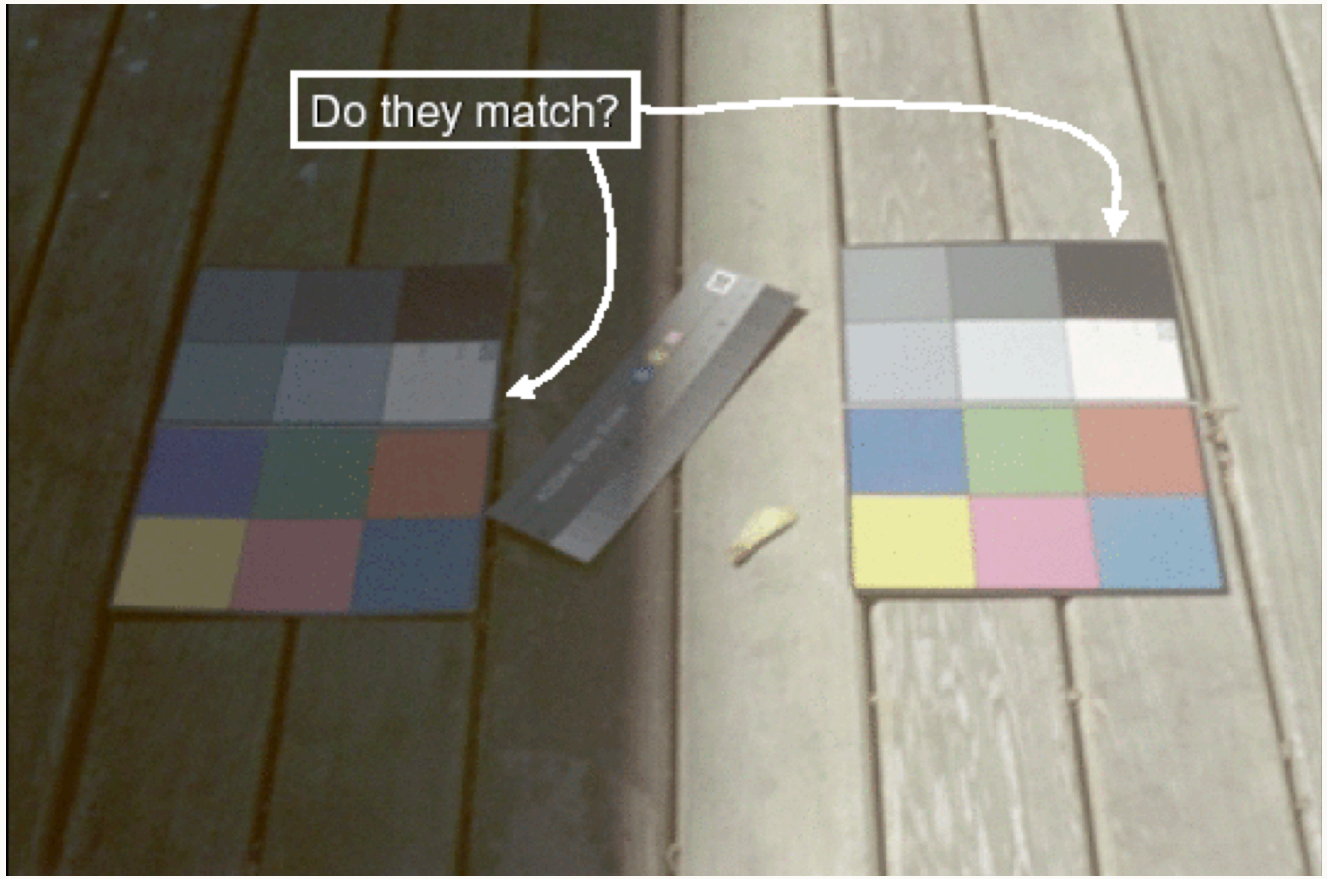
The visual system compares cone responses to support color perception.

Changing hue can also change perceived lightness. A colormap needs both dimensions checked.

How people perceive and interpret color

How viewing conditions, culture, and comparison tasks shape color judgments.

Illumination changes how a color appears



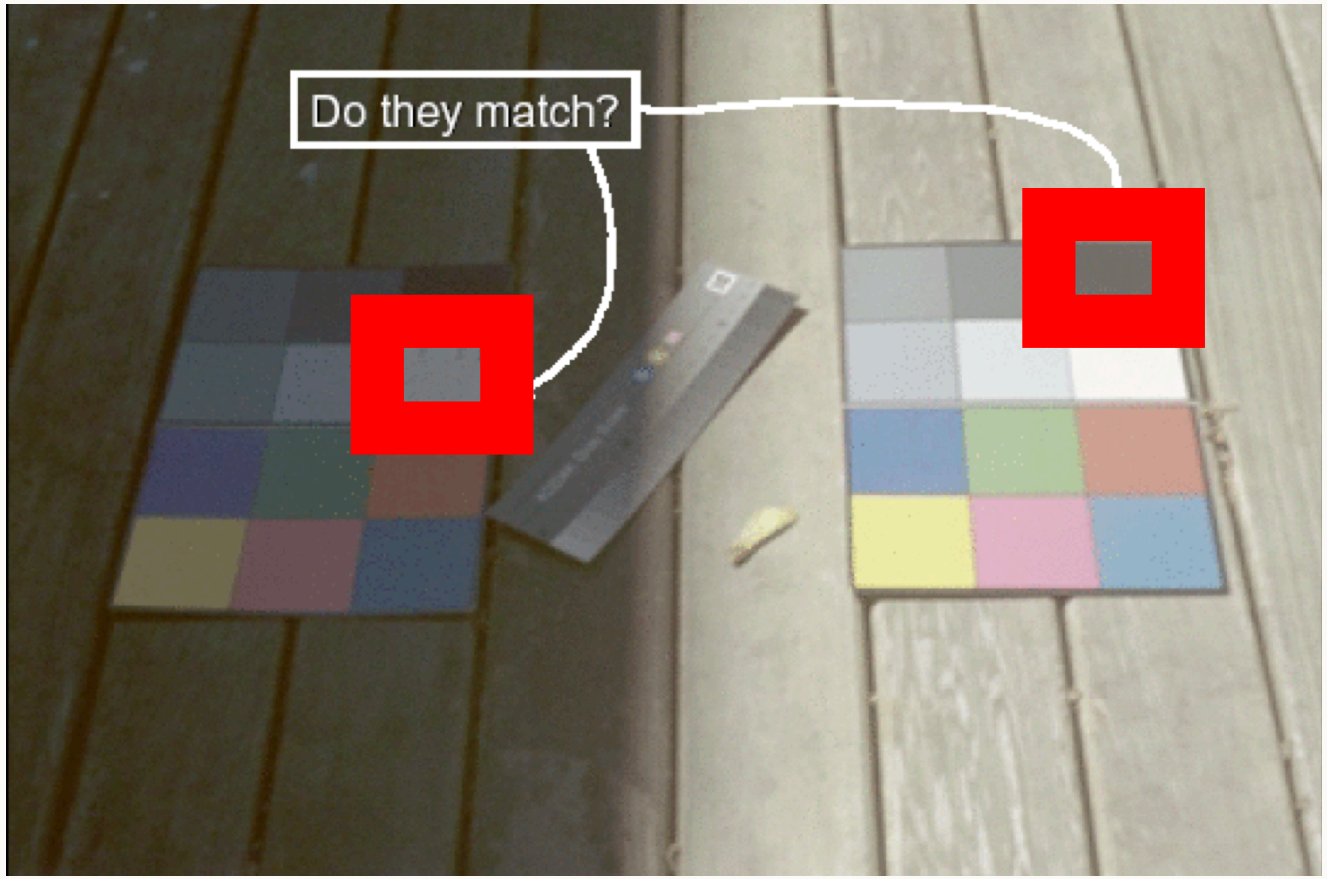
Color constancy is the tendency for surface colors to appear relatively stable as illumination changes.

The visual system interprets a patch in relation to the light falling on the scene.

Compare the two indicated patches. Similar image values can produce different impressions of surface color.

Reading: Robert Simmon · NASA · The “Perfect” Palette

A shared surround changes the comparison



The red frames give the two patches a more similar **immediate surround**.

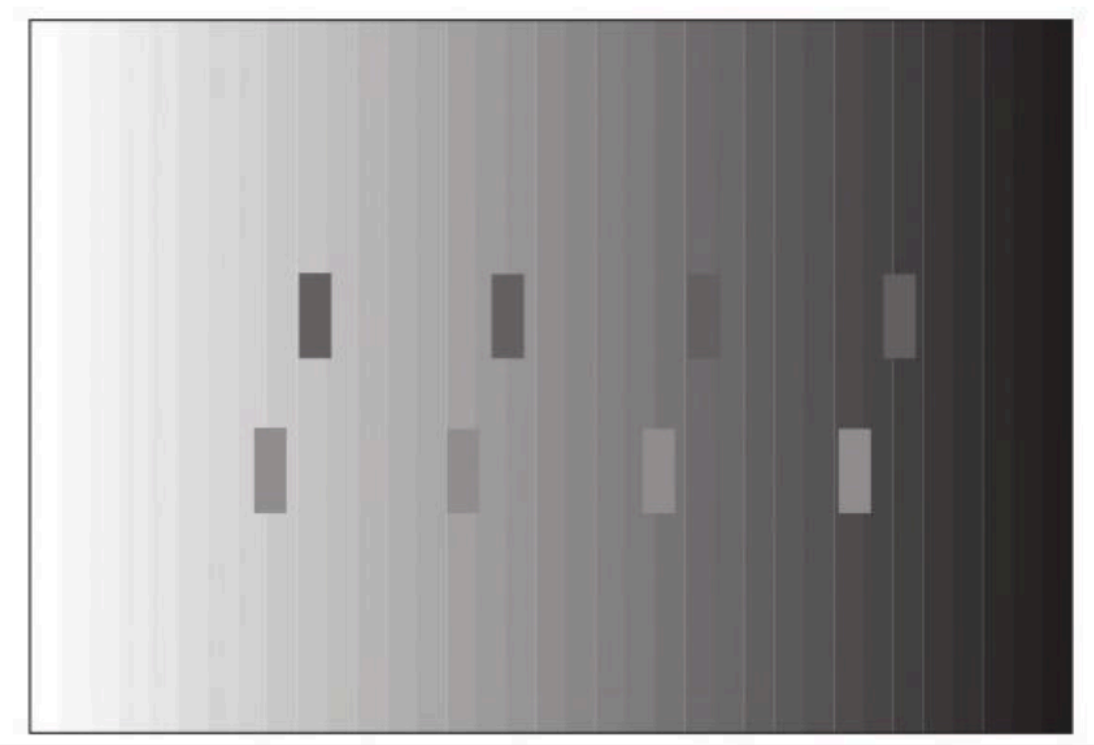
Compare their appearance with the previous page.

The frames change context. They do not change the pixel values inside the patches or the light in the photographed scene.

Reading: Robert Simmon · NASA · The “Perfect” Palette

Surroundings change apparent lightness

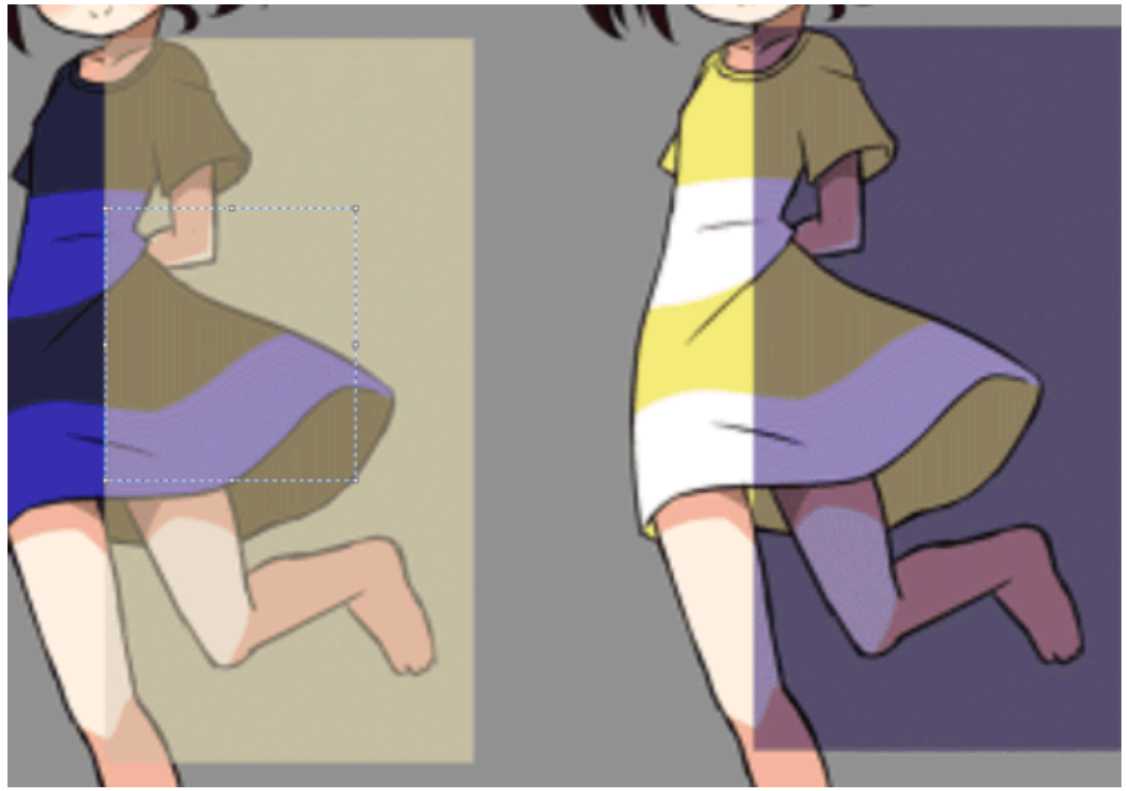
Reference grays



Compare the repeated squares within each row.

Reading: Robert Simmon · NASA · Simultaneous contrast

The dress shows disagreement about illumination

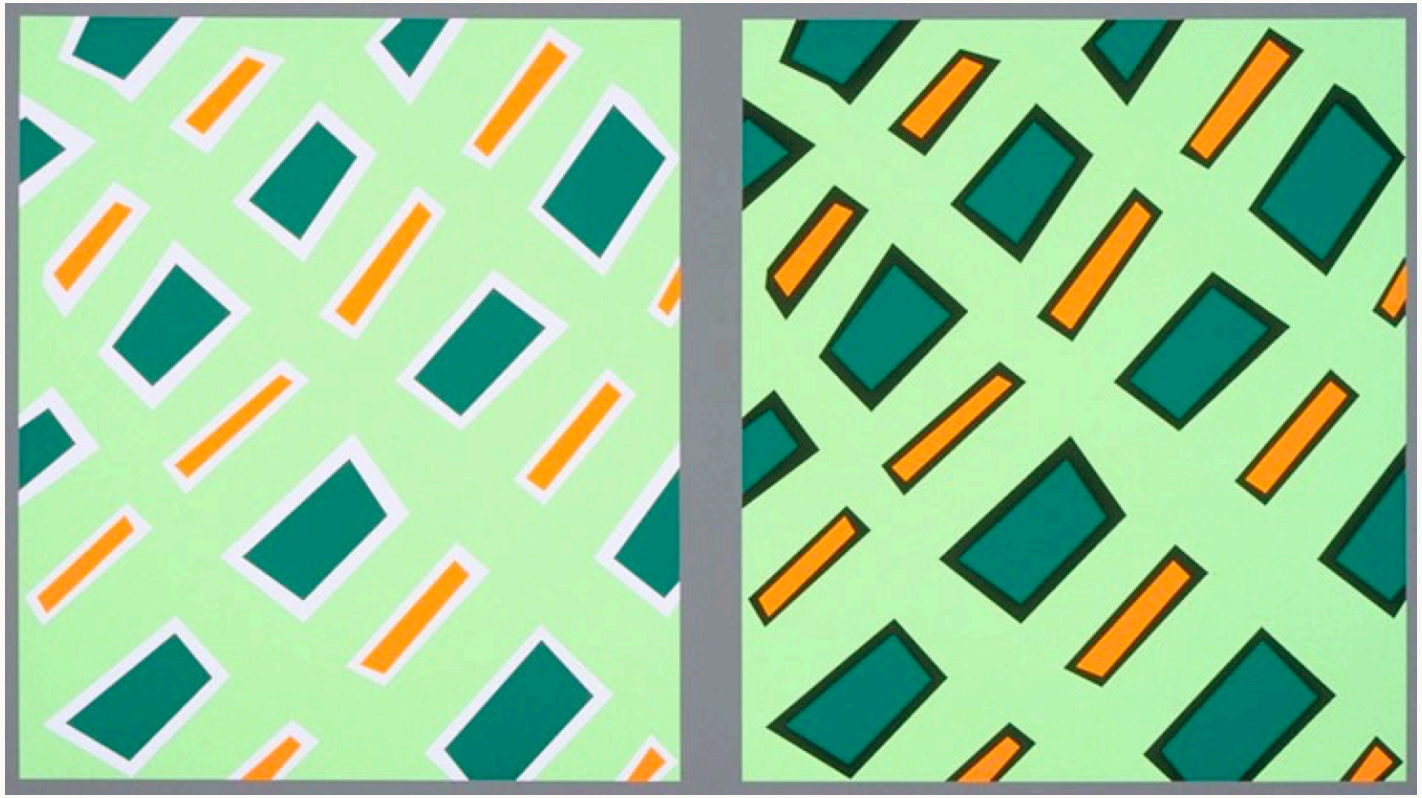


Blue and black, or white and gold? Different assumptions about illumination can change the perceived surface colors.

Illustration link · Background: The Dress

Adjacent colors can alter an entire pattern

The **Bezold effect** is a change in the appearance of a patterned color caused by changing neighboring colors.

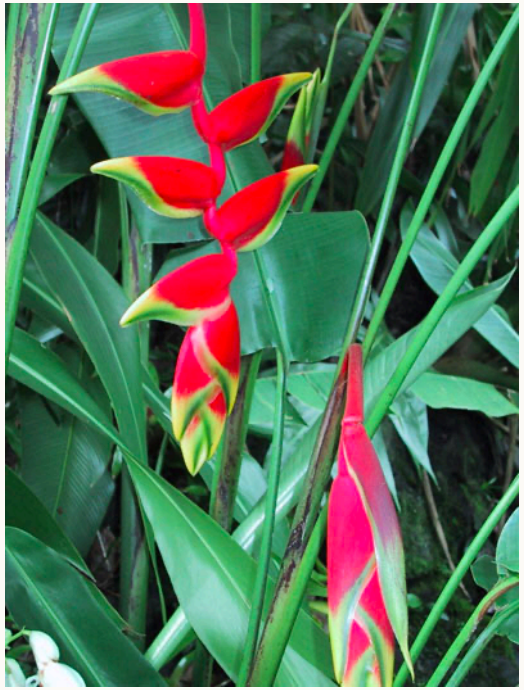


Compare the colored regions when the surrounding outlines switch from light to dark.

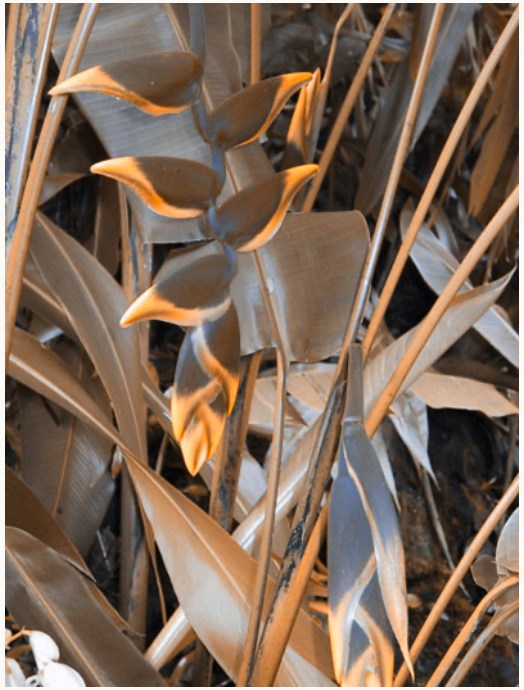
Bezold effect

Color-vision differences change hue distinctions

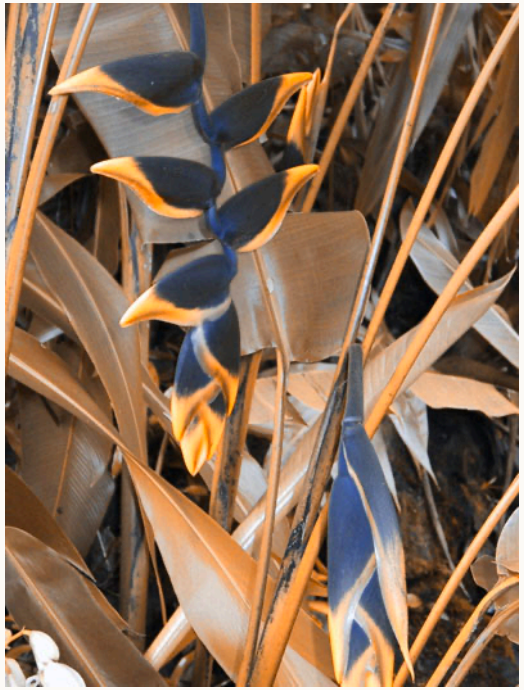
Color vision deficiency changes the ability to distinguish some colors. Different types affect different color combinations.



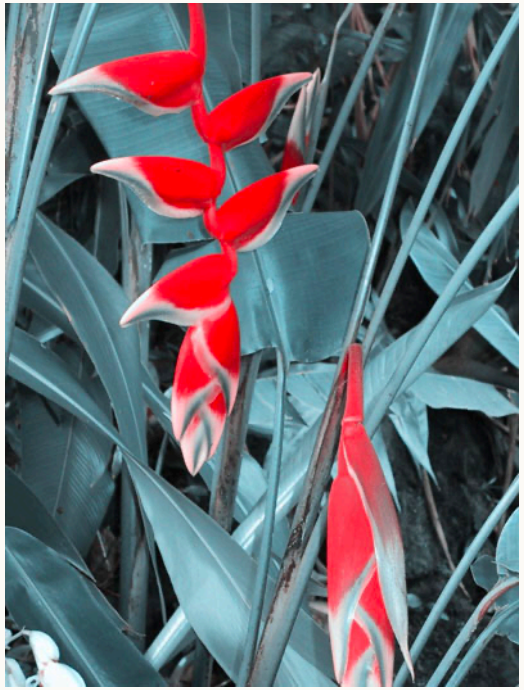
Original



Deuteranopia



Protanopia



Tritanopia

Reading: National Eye Institute · Types of Color Vision Deficiency

Culture shapes the meanings of colors

Cultural color associations are meanings learned through shared symbols, traditions, and social practices.

In many western cultures, yellow is associated with happiness and joy.

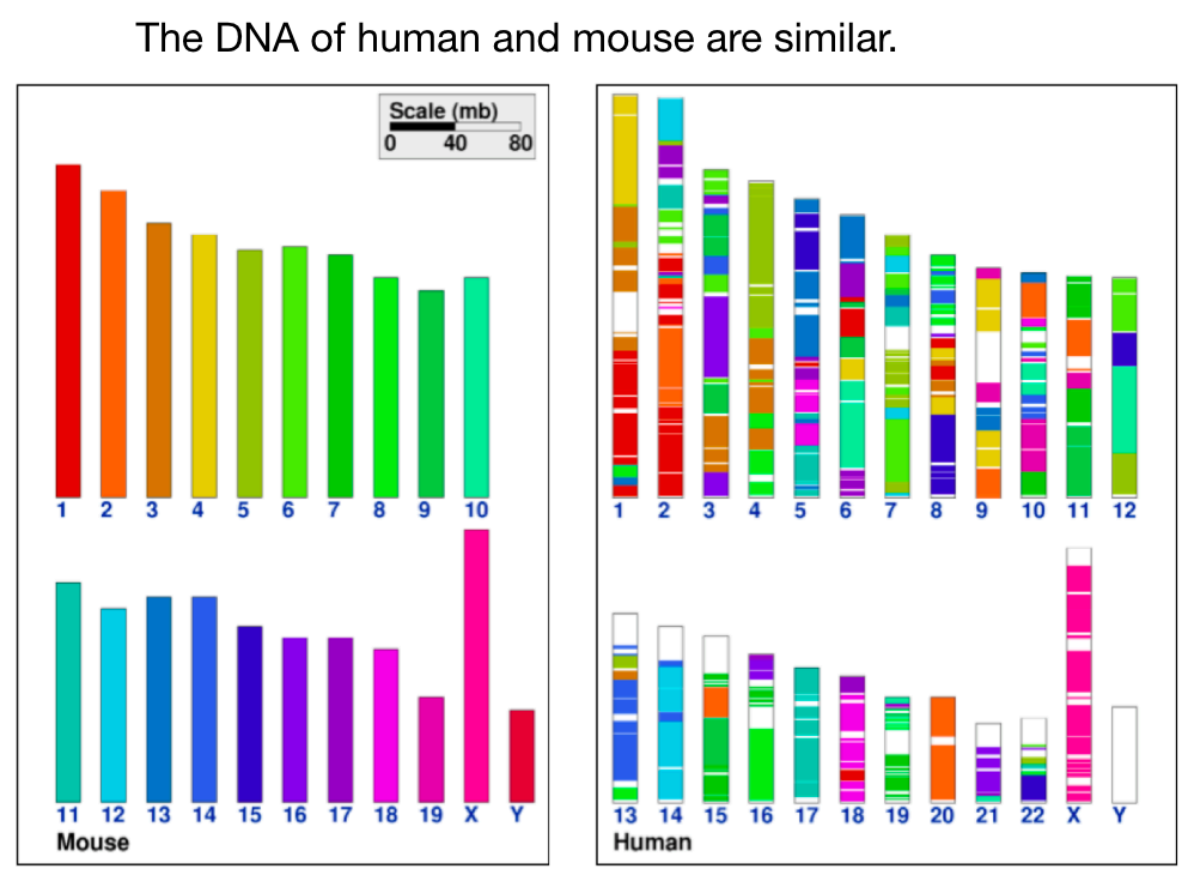
In east Asian cultures, yellow is associated with caution, e.g., in China, yellow robes symbolized imperial status and authority.

For visualization, check the audience's cultural associations and explain the intended color meaning.

Reading: Smithsonian · Smiley history · Qing court dress

Color capacity depends on the comparison task

The number of distinguishable colors depends on **mark size, spacing, contrast, palette, and task**.



Adjacent comparison: large, nearby bars in the mouse key make small color differences easier to see.

Cross-view matching: small, separated segments in the human diagram make remembering and matching hues harder.

Design response: highlight corresponding regions or add labels to reduce reliance on color memory.

Optional exercise: Compare Colors

Color design must account for the audience

Perceptual differences: contrast, mark size, viewing conditions, and comparison layout change what people can distinguish.

Color-vision differences: some hue distinctions become difficult for part of the audience.

Learned associations: familiar meanings vary by setting and community.

Design response: begin with an established palette, explain the mapping, and check it with the intended task and audience.

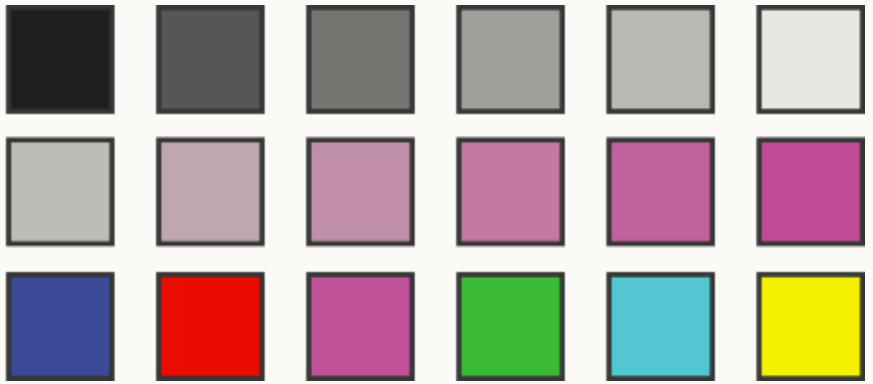
NASA · Color and perception · National Eye Institute · Color vision

Color has three controllable dimensions

Lightness is perceived brightness relative to a similarly illuminated white. It can express order.

Saturation describes colorfulness relative to brightness. It is related to, but distinct from, chroma.

Hue distinguishes appearances such as red, green, and blue. It often expresses category identity.



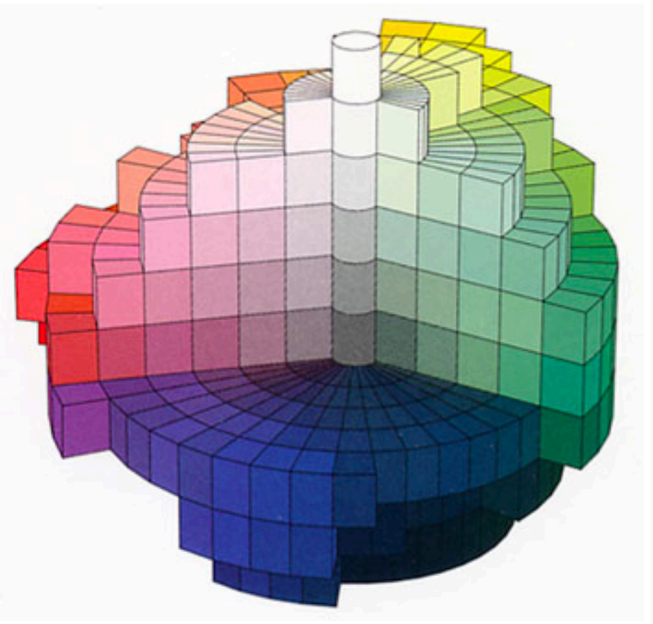
Color models provide different coordinates

RGB specifies the red, green, and blue components used by a display.

HSL / HSV reorganize RGB into hue, saturation, and lightness or value. Equal HSL or HSV steps do not guarantee equally visible differences.

CIELAB uses L* for lightness, a* for a green–red axis, and b* for a blue–yellow axis. It aims for approximate perceptual uniformity.

Perceptual uniformity means that equal distances in a color space correspond approximately to equally visible differences.



Reading: Matplotlib · Choosing Colormaps

Colormaps

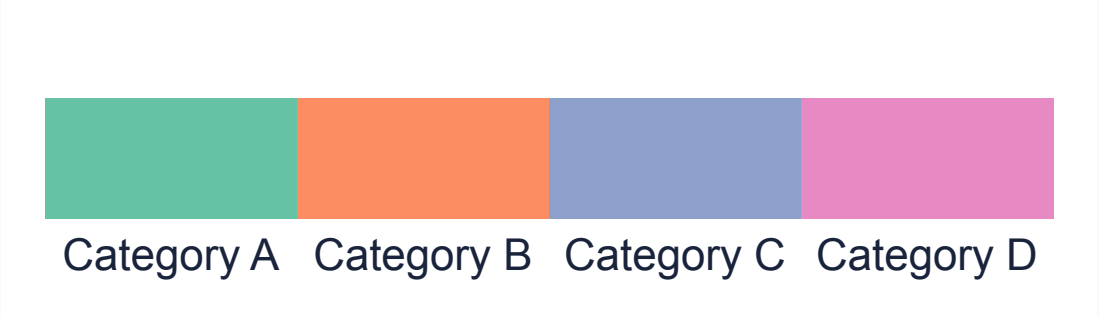
Matching data relationships to visible color relationships.

A colormap assigns data values to colors

A **colormap** is a rule that assigns a color to each data value or combination of values in a defined domain.

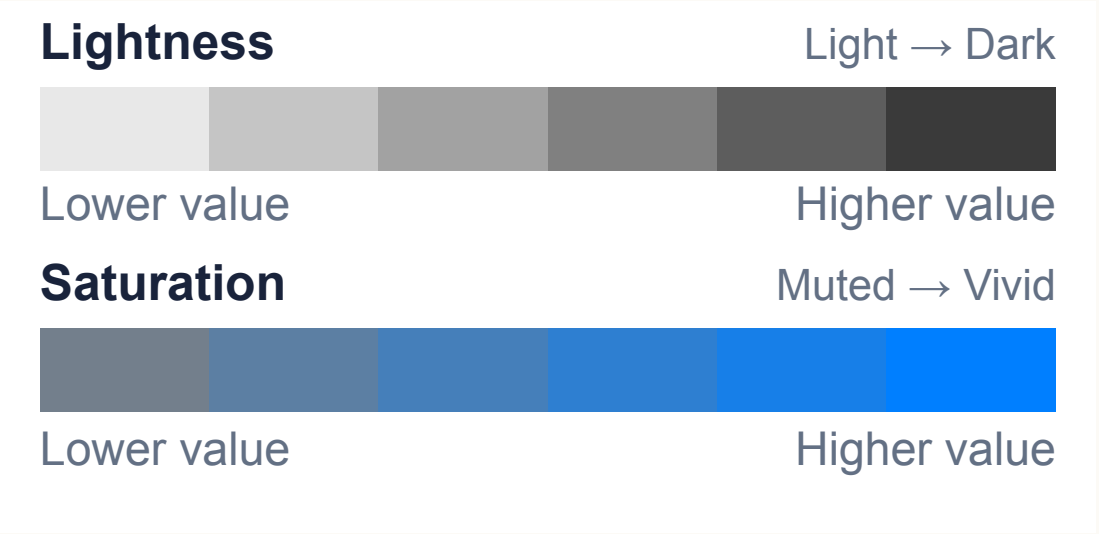
Category identity

Hue primarily expresses **category identity**.



Ordered attributes

Lightness and saturation can encode **order**.



Data values → Mapping rule → Colors

ColorBrewer · Set2 · Ordered ramps: conceptual illustrations

Design Effective Colormaps

We choose (or design) a colormap based on the properties of the data:

- **Categorical / Ordered** — depends on the type of attribute
- **Continuous / Segmented** — depends on the range of values
- **Univariate / Bivariate** — depends on the number of attributes encoded

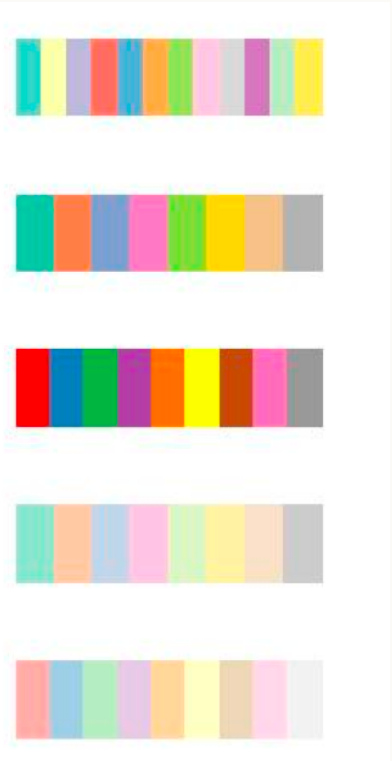
Note: these categories are not mutually exclusive — for example, a colormap can be both categorical and segmented.

ColorBrewer · Scheme types · Sarah Battersby · Bivariate maps

Categorical palettes distinguish group identities

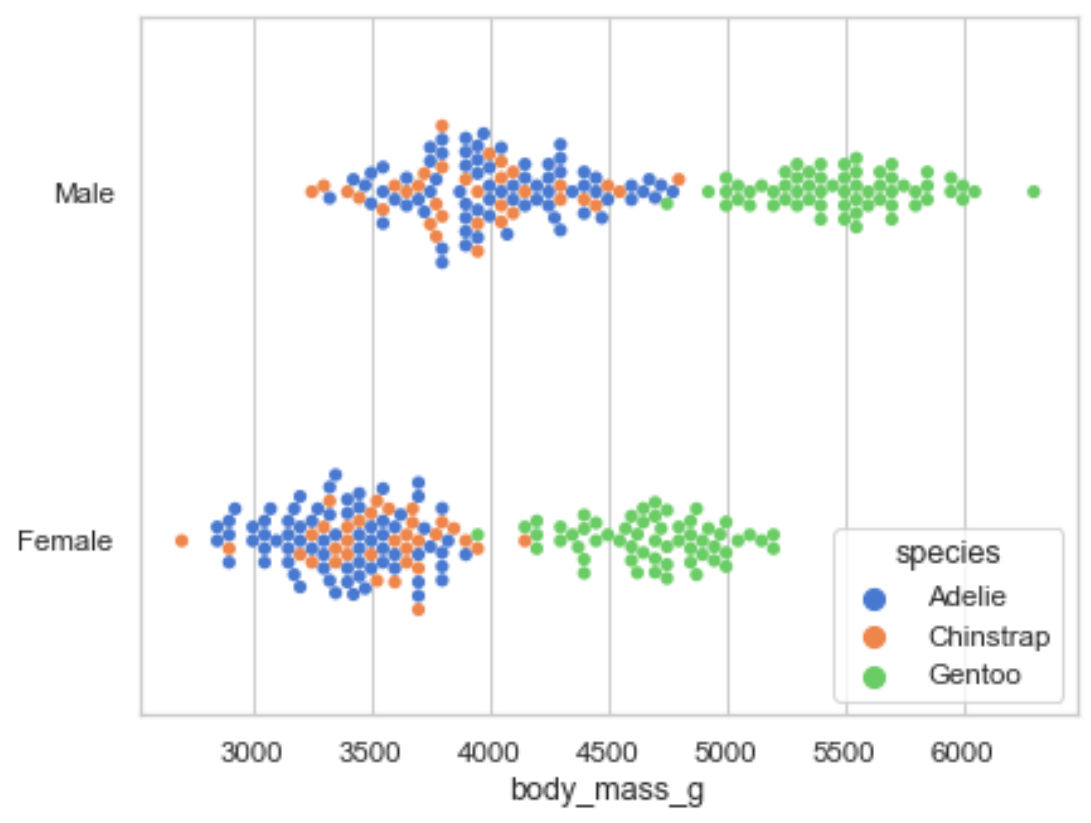
A **categorical colormap** assigns distinct colors to groups with no inherent numeric order.

- Use hue to encode categories or groupings.
- Ensure distinguishability — typically effective for fewer than 12 hues.
- Most visualization libraries provide built-in categorical palettes — use them when possible.
  - Matplotlib: **tab10** and **Set2**.

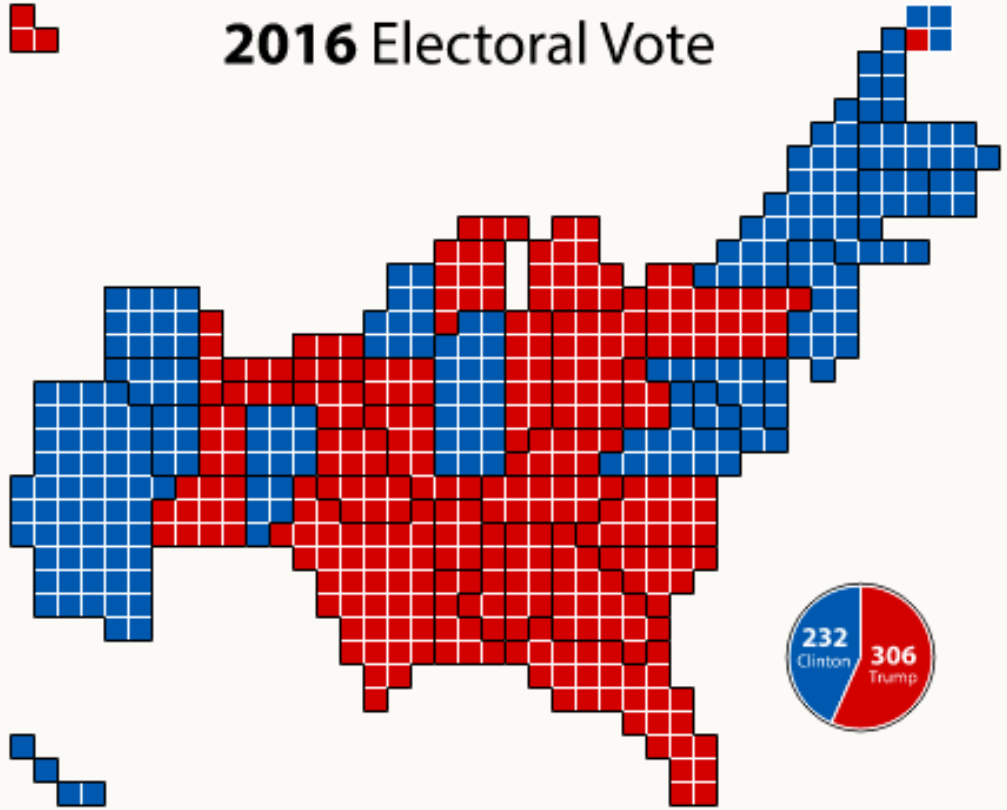


ColorBrewer · Qualitative schemes · Matplotlib · Colormap reference

Categories can identify species or political groups



Penguin species: Adelie, Chinstrap, and Gentoo.



Political identity in a historical election cartogram.

Left data labels: Palmer penguin species · Right: 2016 electoral-vote illustration

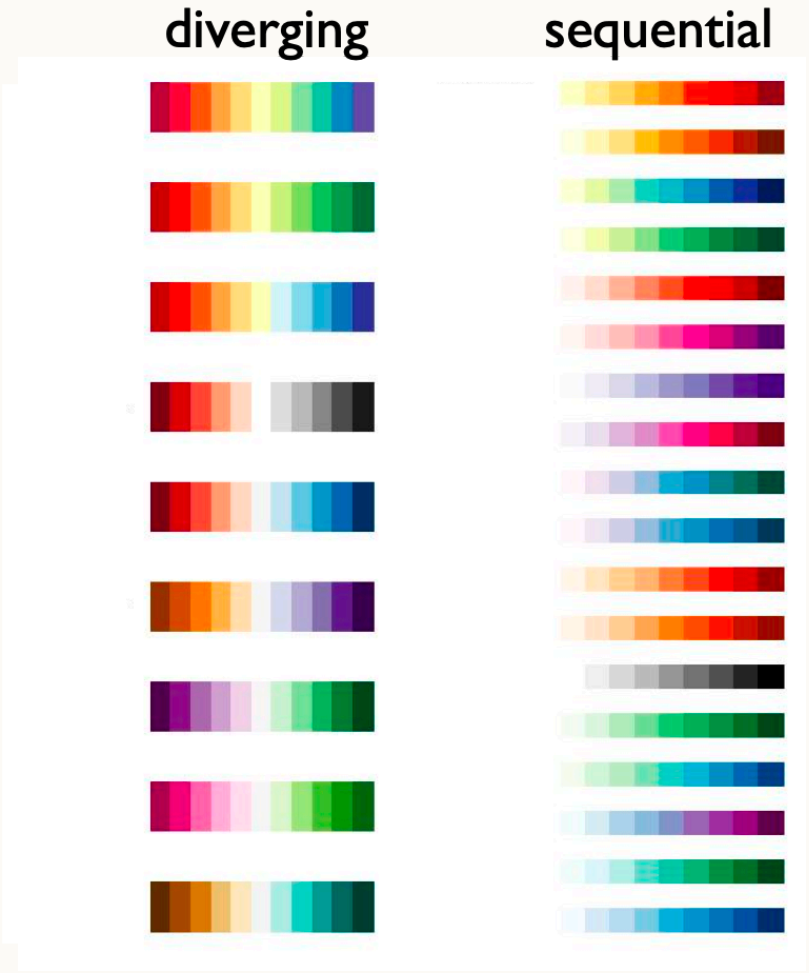
Ordered palettes need a readable progression

An **ordered colormap** represents ordinal or quantitative values through a visible color progression.

Sequential: one direction from low to high.

Diverging: two directions meeting at a meaningful reference.

Check lightness and perceptual differences so accidental visual jumps do not imply unsupported data boundaries.

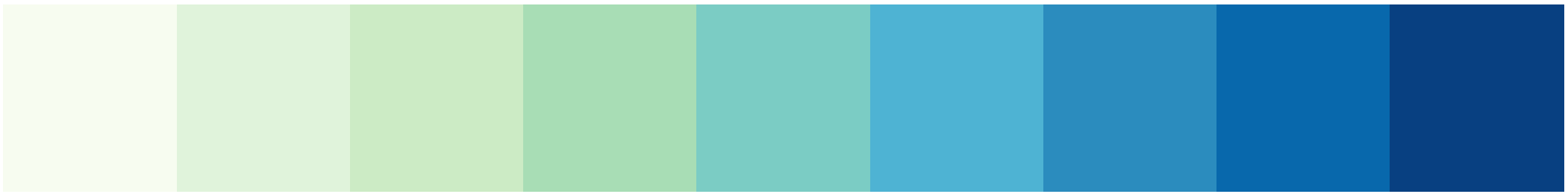


ColorBrewer · Sequential and diverging schemes · Matplotlib · Perceptual properties

Sequential palettes express low-to-high order

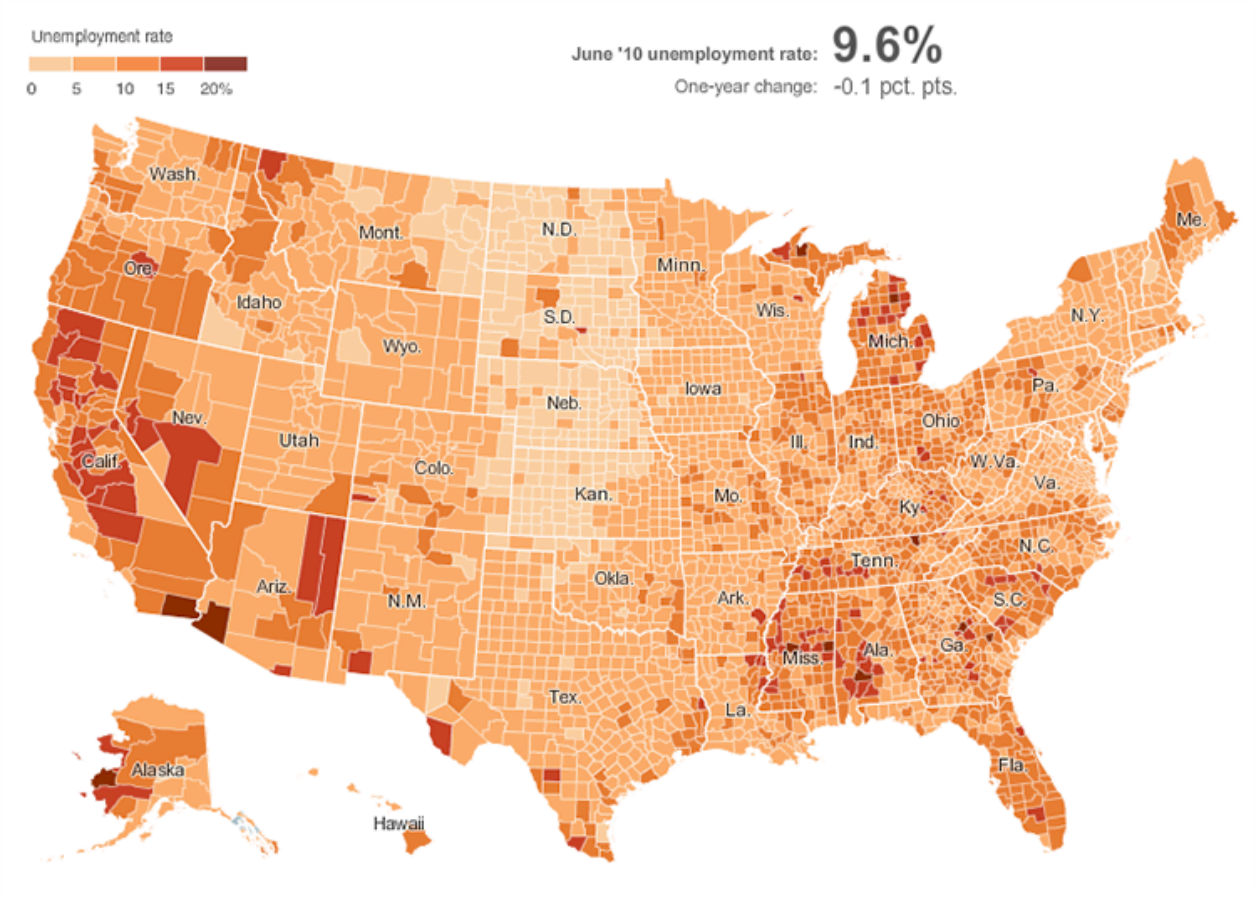
A **sequential colormap** represents an ordered variable through a progression from low to high.

A monotonic lightness progression provides a visible ordering cue. Hue may also vary along that progression.



ColorBrewer · Color scheme types · Matplotlib · Choosing Colormaps

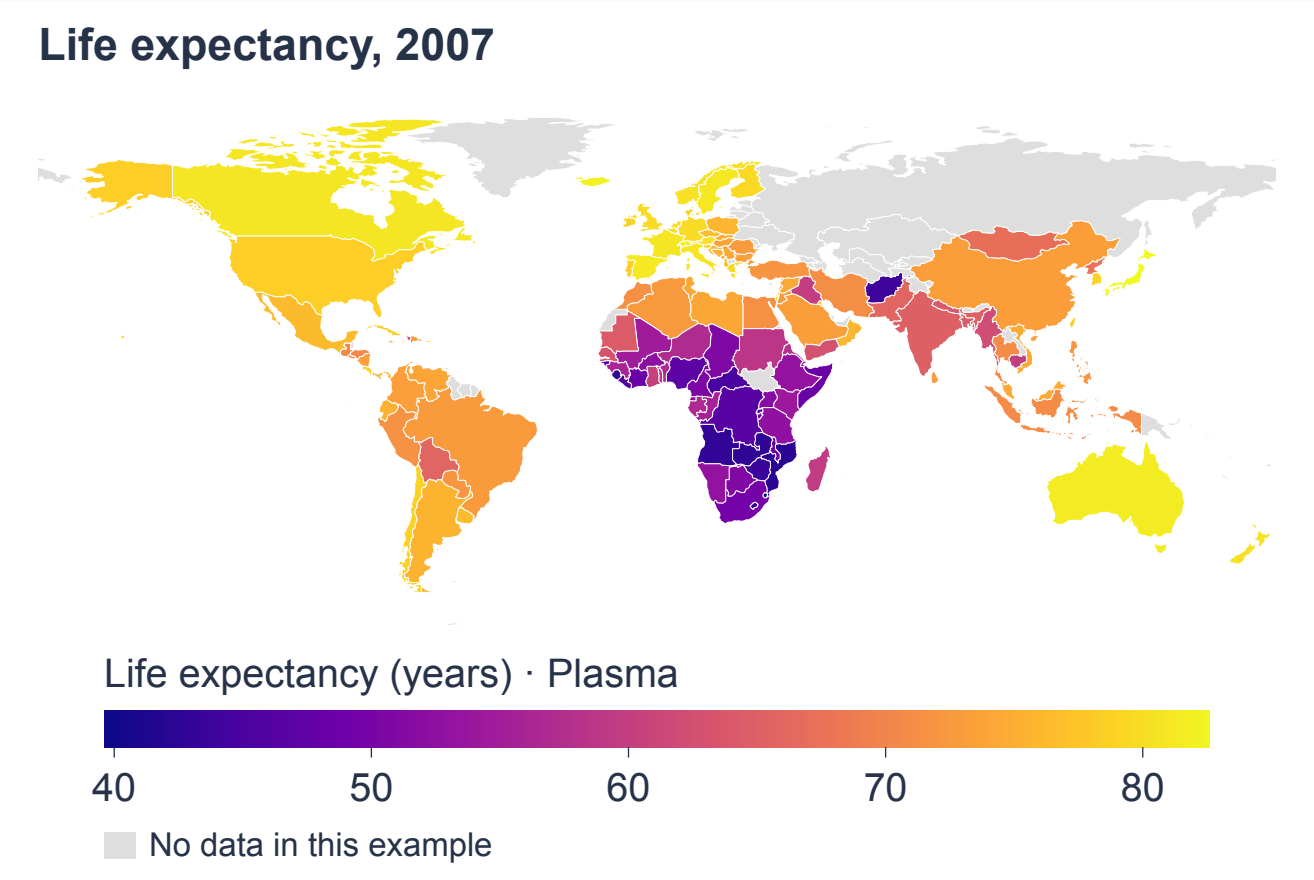
NYT makes higher unemployment appear darker



The legend maps an ordered percentage to a light-to-dark orange progression.

The New York Times · June 2010 view · Image preserved by Joshua Stevens

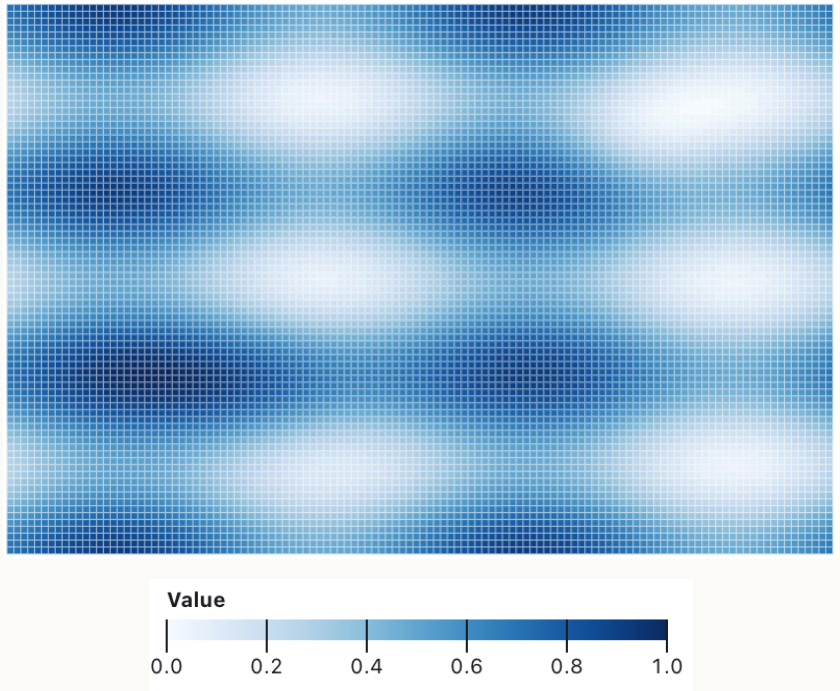
A sequential map may use several hues

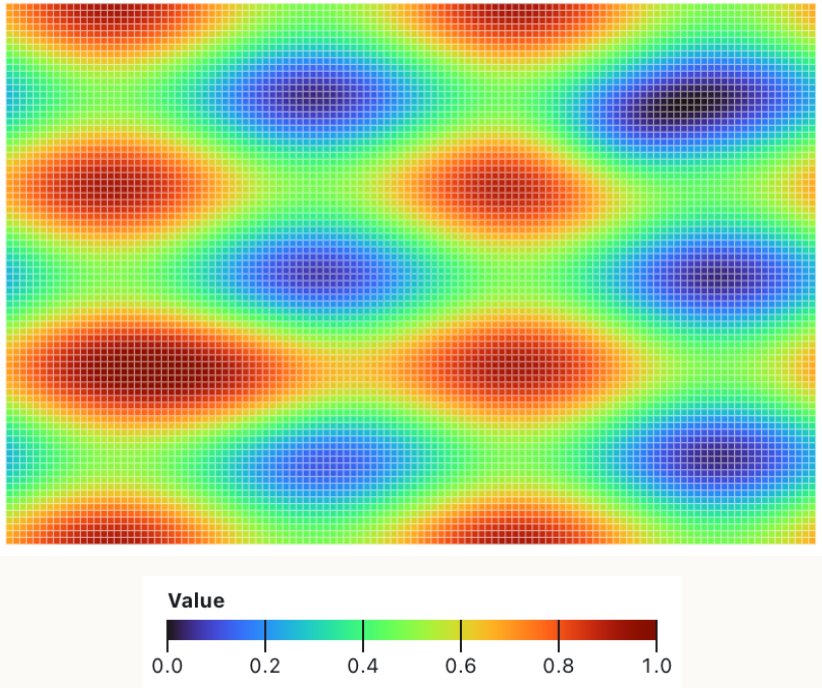


- A **multi-hue** scheme can also work if its lightness progression supports the intended order.
- Lightness should change consistently. Hue (wavelength) order alone does not establish numeric order.

Adapted from Plotly · Gapminder (2007) · Plasma colormap

More hues can change which details attract attention





Single hue: a light-to-dark progression.

Multiple hues: stronger local color changes.

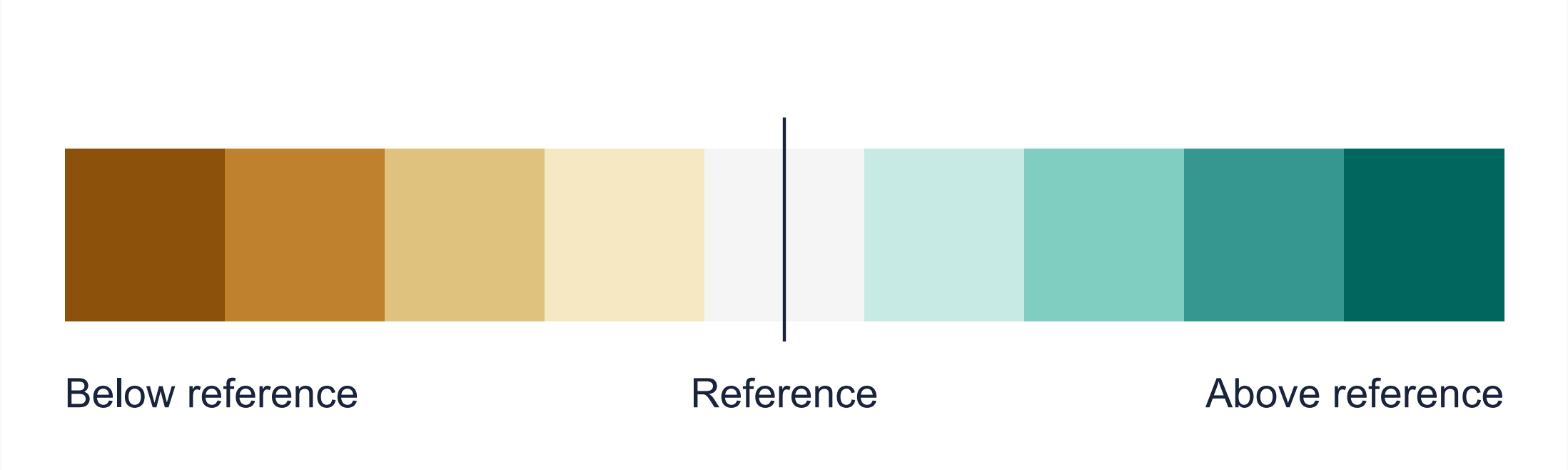
Extra hue changes may reveal detail, but can also introduce apparent boundaries. Check that the visual emphasis serves the task.

Reading: Matplotlib · Colormap selection

Diverging palettes organize values around a reference

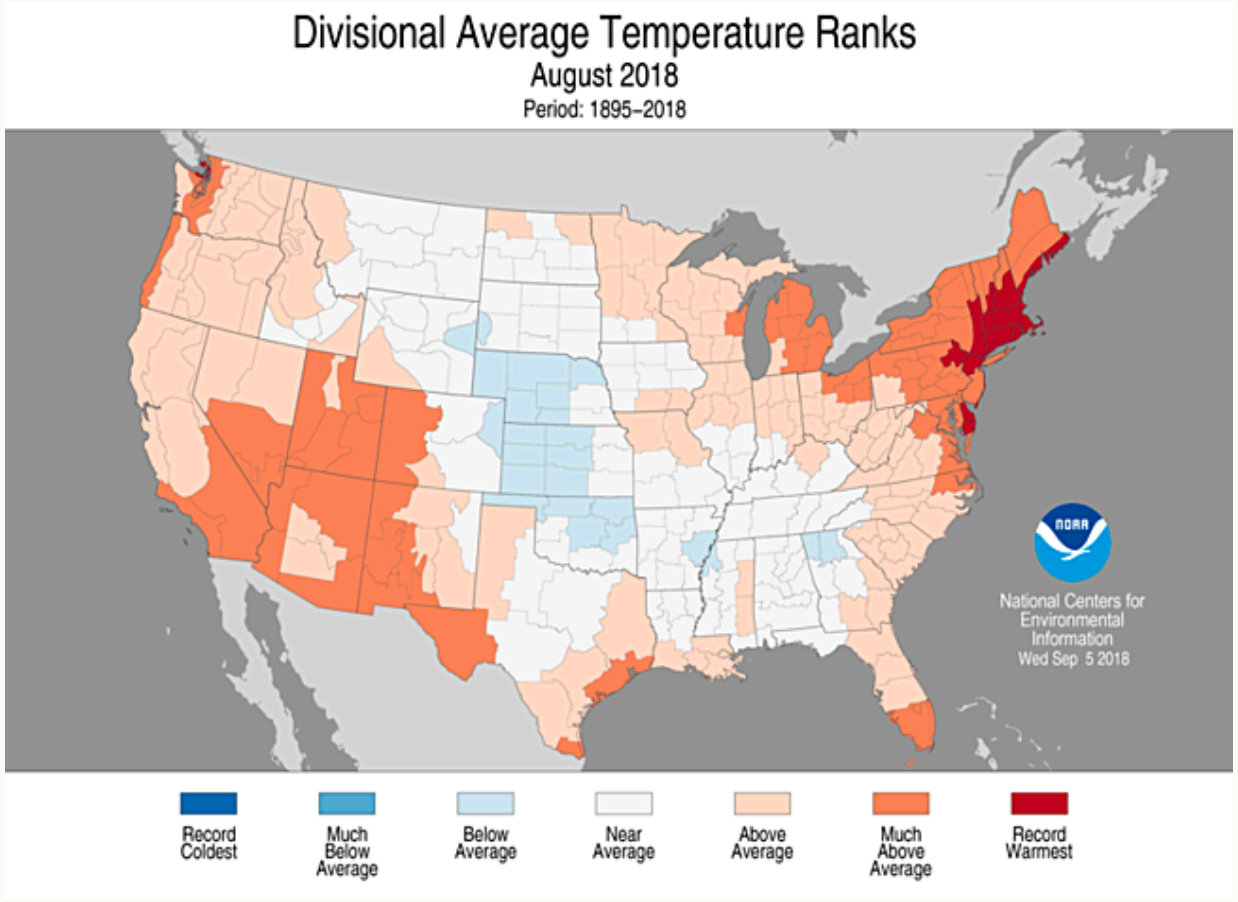
A **diverging colormap** uses two ordered color progressions that meet at a meaningful reference value or reference range.

The two sides indicate different directions from the reference, such as loss and gain or colder and warmer than expected.



ColorBrewer · Color scheme types

NOAA gives “near average” a neutral color



Blue and red distinguish colder and warmer ranks; the reference is the “Near average” class.

NOAA / NCEI · U.S. climate maps · August 2018

Continuous and segmented mappings support different readings

Segmented: one color per category or numeric interval. Binning simplifies reading but hides differences within each interval.

Continuous: color varies smoothly with value, preserving finer variation.

Both sequential and diverging schemes can be continuous or segmented.

ColorBrewer · Schemes · Matplotlib · Colormaps

How will values become colors?

A good palette can still produce a misleading mapping.

Choose the domain and the normalization

The **domain** is the set or range of data values covered by the mapping.

Normalization transforms numeric values into positions along a colormap, often from 0 to 1.

With linear normalization:
t = (x – minimum) / (maximum – minimum)

With logarithmic normalization:
t = log(x / minimum) / log(maximum / minimum)
Requires x > 0 and 0 < minimum < maximum.

Three separate choices

Domain: which values define the ends?

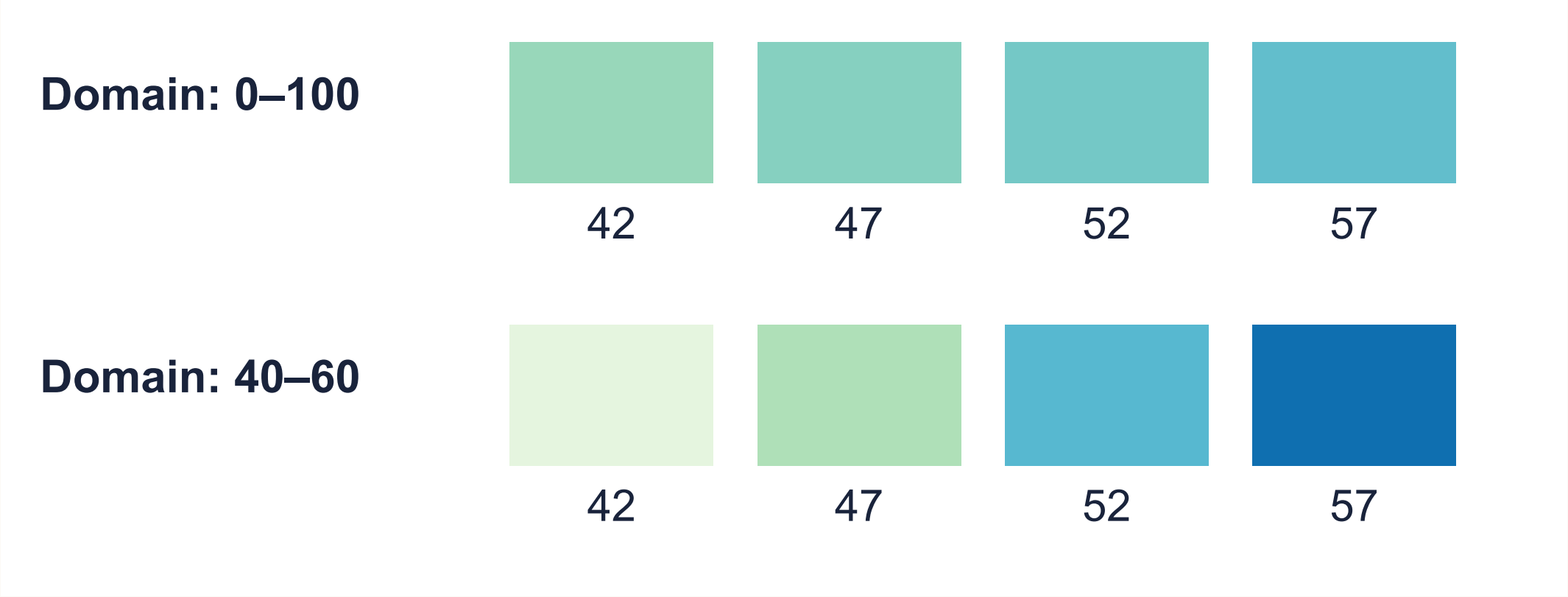
Normalization: how is space allocated between them?

Palette: which color appears at each position?

Specify how out-of-range values are handled.

Matplotlib · Colormap normalization

The domain controls the visible contrast



The values are identical in both rows. A narrower domain makes their differences more visible.

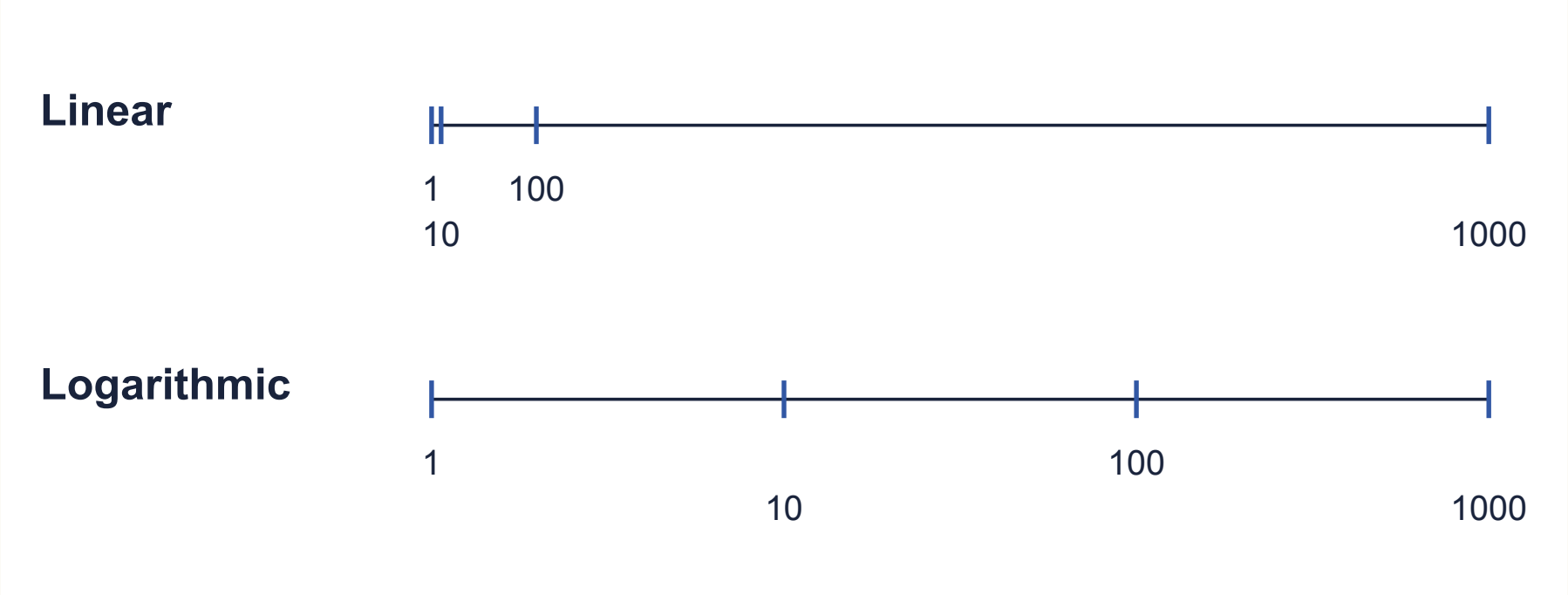
For numerical comparisons across views, keep the **domain, normalization, and palette consistent**.

Conceptual demonstration · illustrative values

Linear and logarithmic scales answer different questions

Linear normalization gives equal numeric differences equal distances along the colormap. Use it to compare **absolute differences** in the original units.

Logarithmic normalization gives equal ratios equal distances. A standard log scale requires positive values. Use it to compare **ratios across orders of magnitude**.

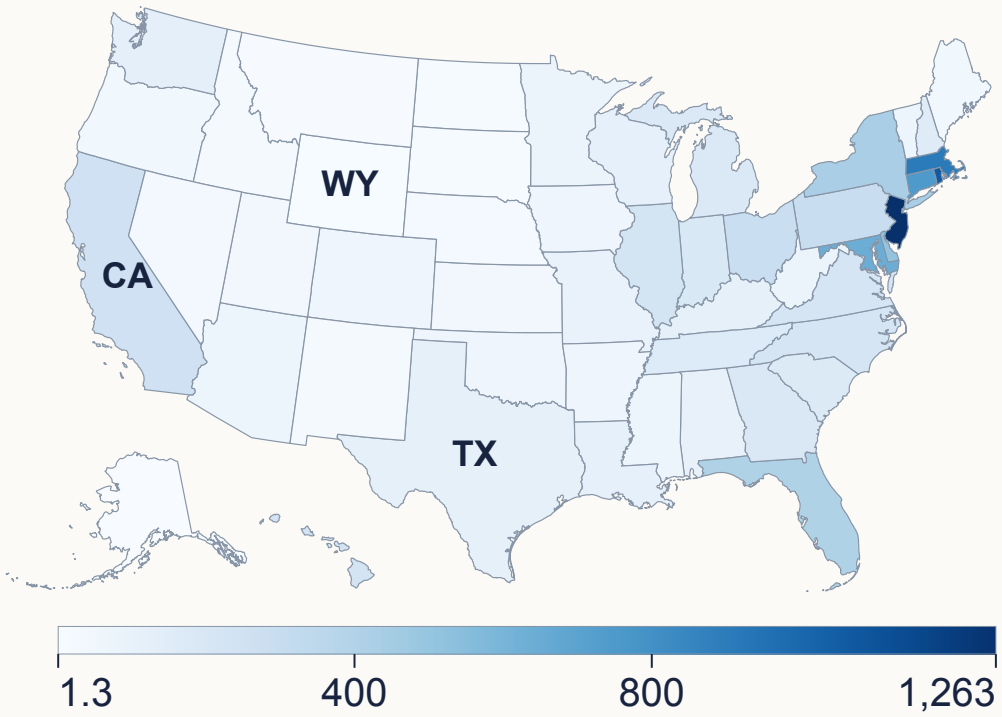


Matplotlib · Colormap normalization · Conceptual demonstration

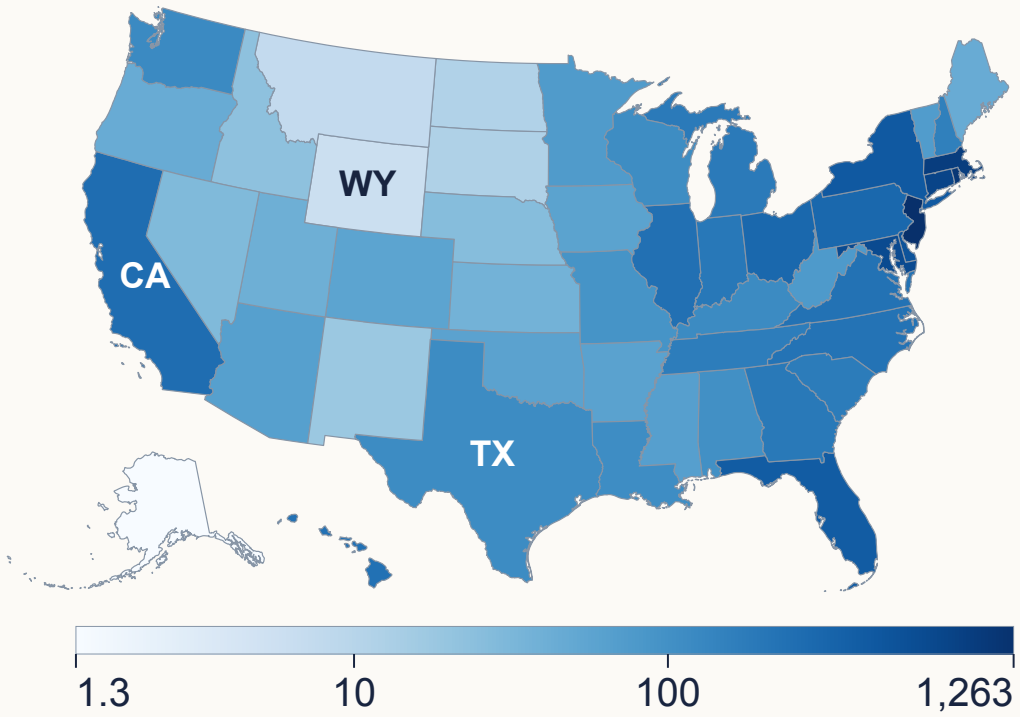
Population density: one dataset, two scales

2020 Census, 50 U.S. states. Same data, palette, and domain (1.3–1,263 people/mi²).

Linear scale



Logarithmic scale



The log scale spreads out lower densities and compresses differences among higher densities.

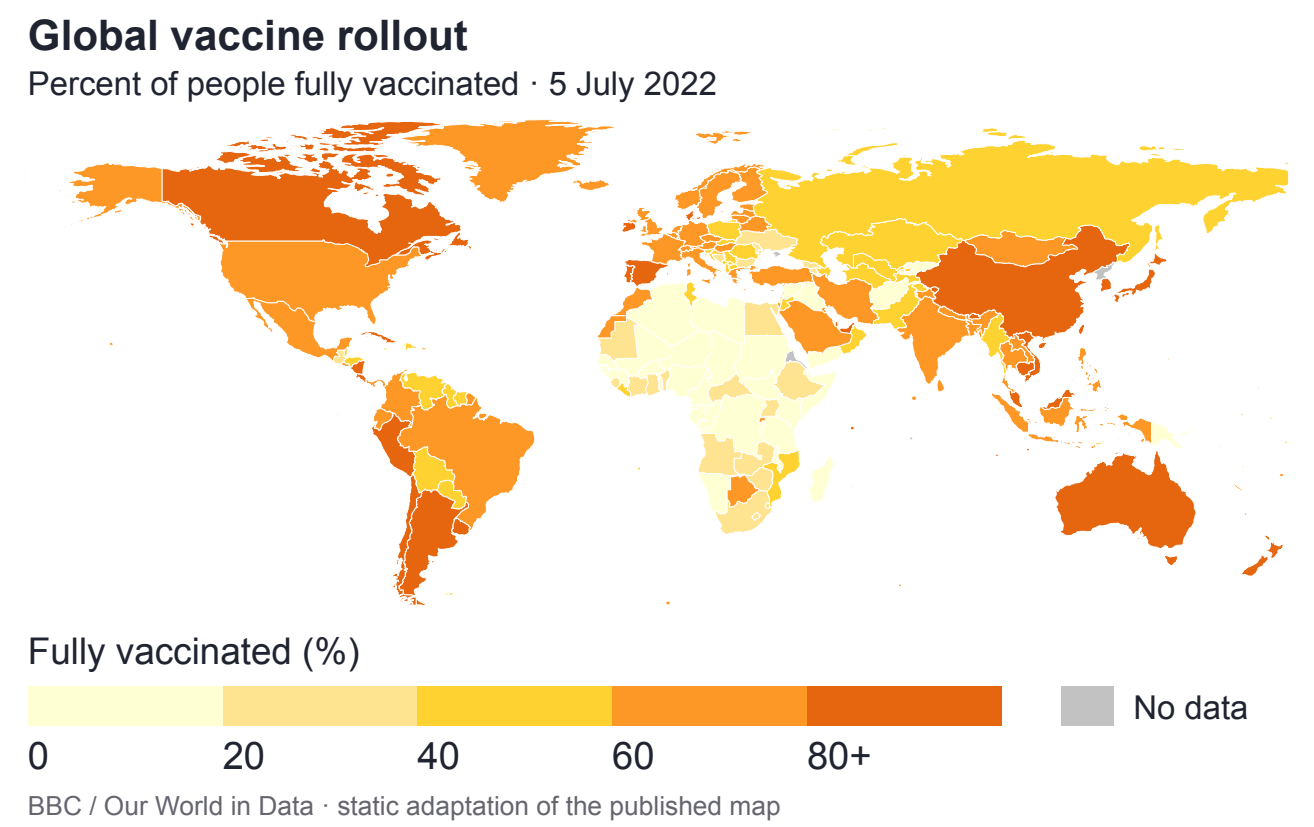
Maps redrawn using U.S. Census Bureau · 2020 population density · Boundaries: us-atlas / U.S. Census Bureau

Class boundaries determine what a difference means

Method	Rule	Interpretation and cost
Equal interval	Equal widths in data units	Easy numeric comparison; skew can leave bins sparse.
Quantile	Similar numbers of observations per bin	Shows relative rank; bin widths may differ greatly.
Meaningful thresholds	Boundaries come from the task	Supports a decision; the thresholds need justification.

Sarah Battersby · Tableau · Bivariate choropleth maps (2018)

Equal interval: vaccination rates



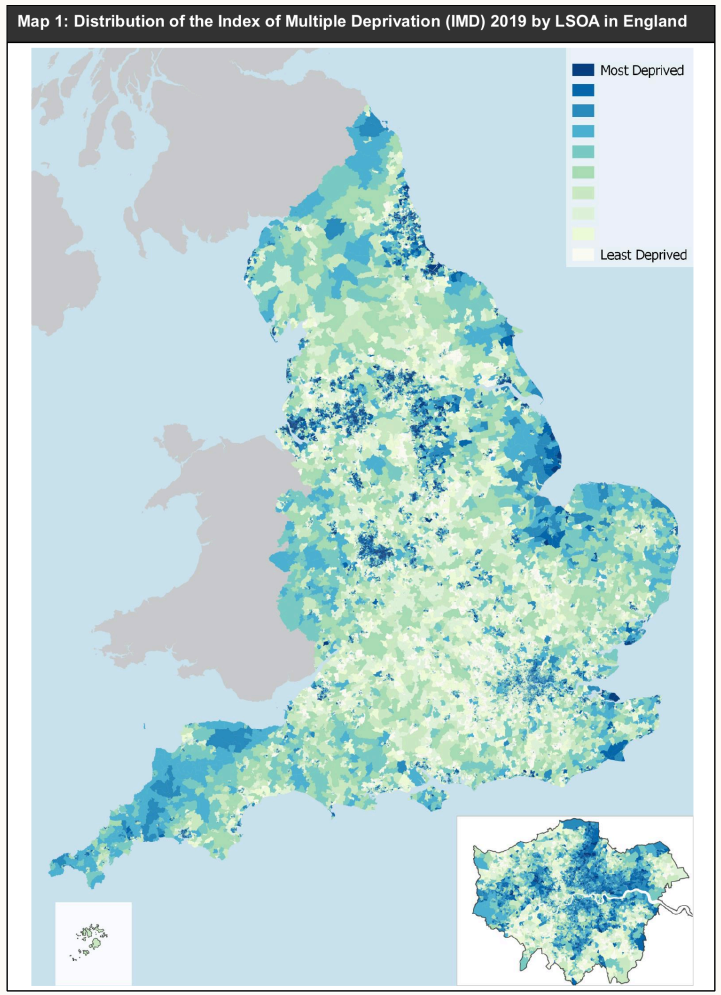
The BBC map uses a fixed step of **20 percentage points** between class boundaries.

The number of countries in each class **need not be equal**.

The highest class is open-ended: **80%+**.

BBC News · Global vaccine rollout · 5 July 2022 · Data: Our World in Data · Static map adaptation

Quantile: equal numbers of neighbourhoods



England's **32,844 neighbourhoods** are ranked by deprivation and divided into **10 groups (deciles)**.

Each color ≈ 10% of neighbourhoods

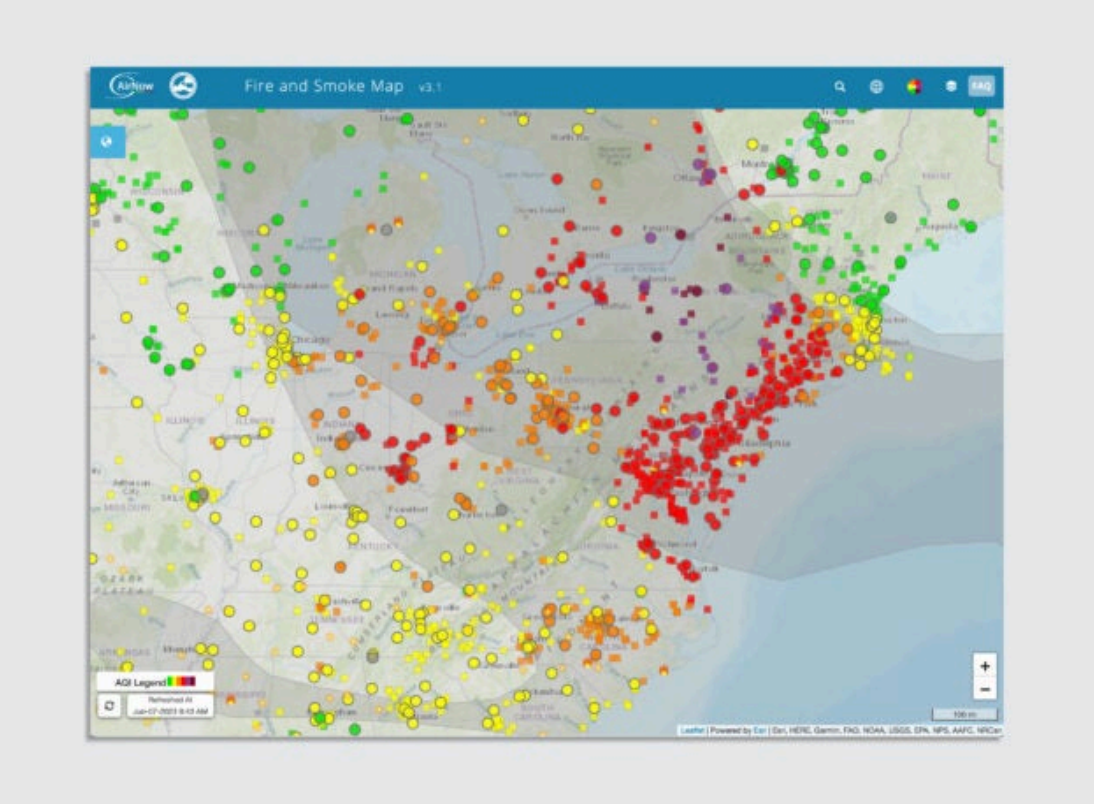
Most deprived 10% — Least deprived 10%

Each group contains about **3,284 neighbourhoods**.

Their score ranges and land areas **need not be equal**.

MHCLG · English Indices of Deprivation 2019 · Map 1 · Original map; enlarged key

Meaningful thresholds: air quality and health



Each monitoring location is colored by its **health-concern category**.

U.S. Air Quality Index (AQI)

0–50	Good
51–100	Moderate
101–150	Unhealthy for sensitive groups
151–200	Unhealthy
201–300	Very unhealthy
301+	Hazardous

Class boundaries follow **health guidance**, rather than equal counts of locations.

Fast Company · Adele Peters · 7 June 2023 · Map: AirNow · U.S. EPA / AirNow · AQI categories

A legend is part of the mapping

Make the numeric meaning recoverable

State the **variable, unit, and time period**.

Show the **domain, direction, ticks or class boundaries**, and any meaningful reference.

Distinguish **zero** from **missing and not applicable** values. Explain any excluded or out-of-range values.

Check the actual reading task

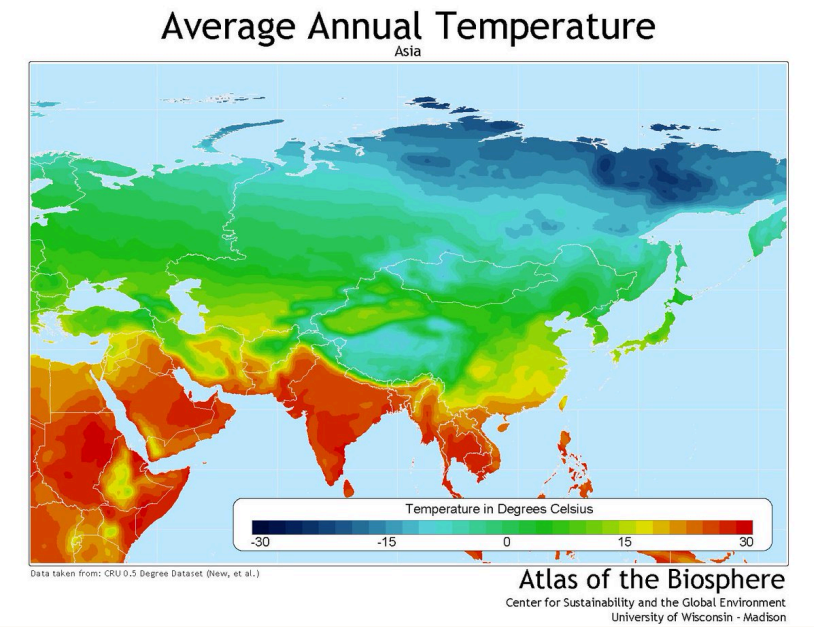
Can the viewer tell whether a color means 5–10%, the bottom 20% of counties, or a value below a target?

If an exact value matters, add a label, table, or interactive readout.

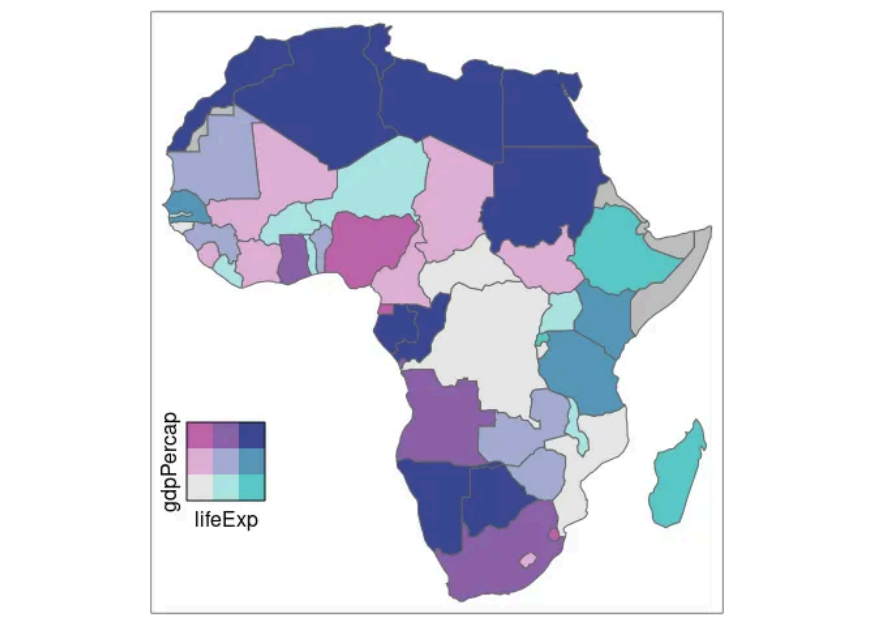
Color alone supports limited numerical precision.

A colormap can encode one value or a pair

A **univariate colormap** assigns colors based on one variable. A **bivariate colormap** assigns colors based on a pair of variables.



One variable: temperature.

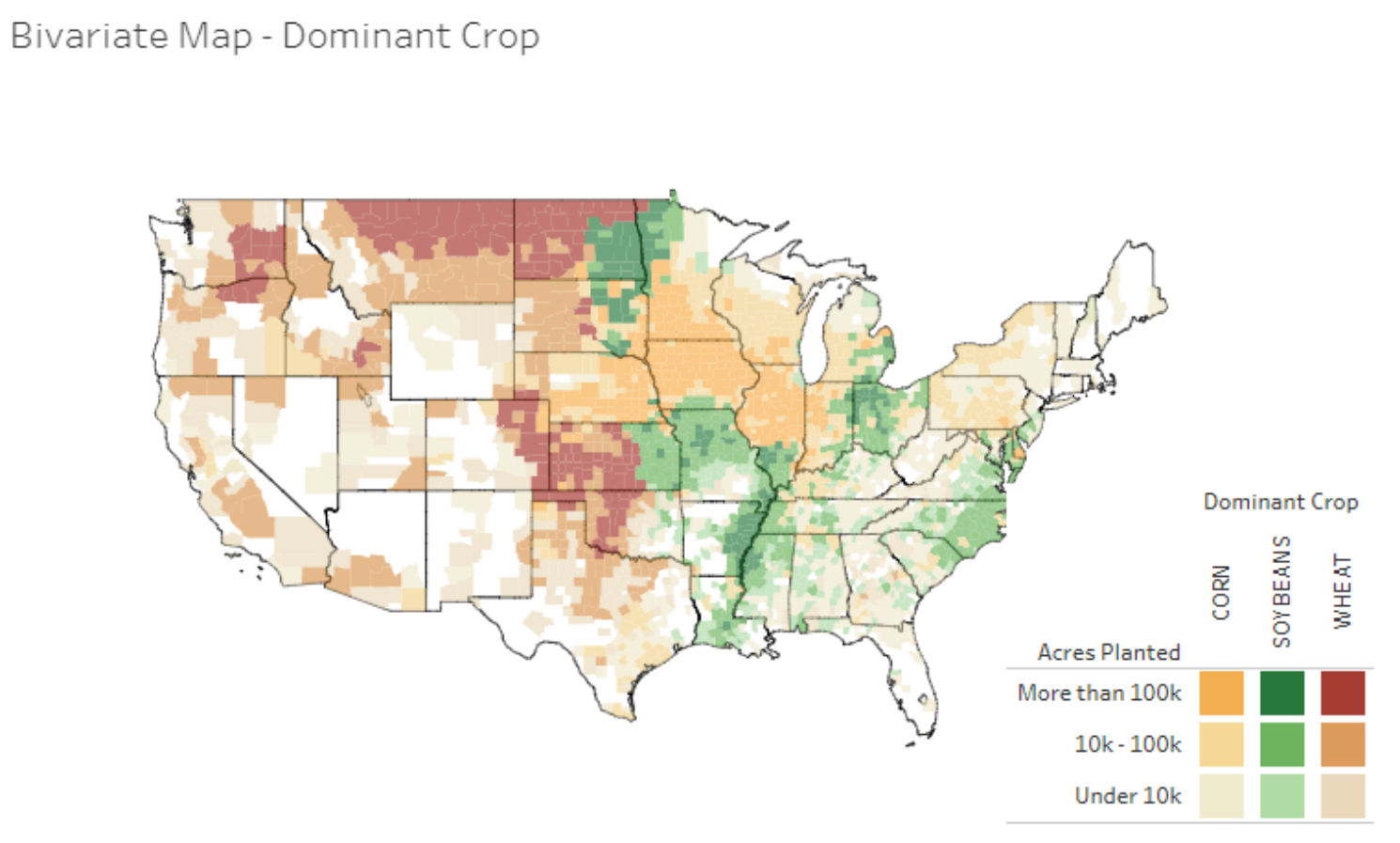


Two variables: GDP per capita and life expectancy.

A bivariate legend must show how both variables contribute to each color.

Temperature map: Atlas of the Biosphere · University of Wisconsin–Madison · Reading: Sarah Battersby · Bivariate maps

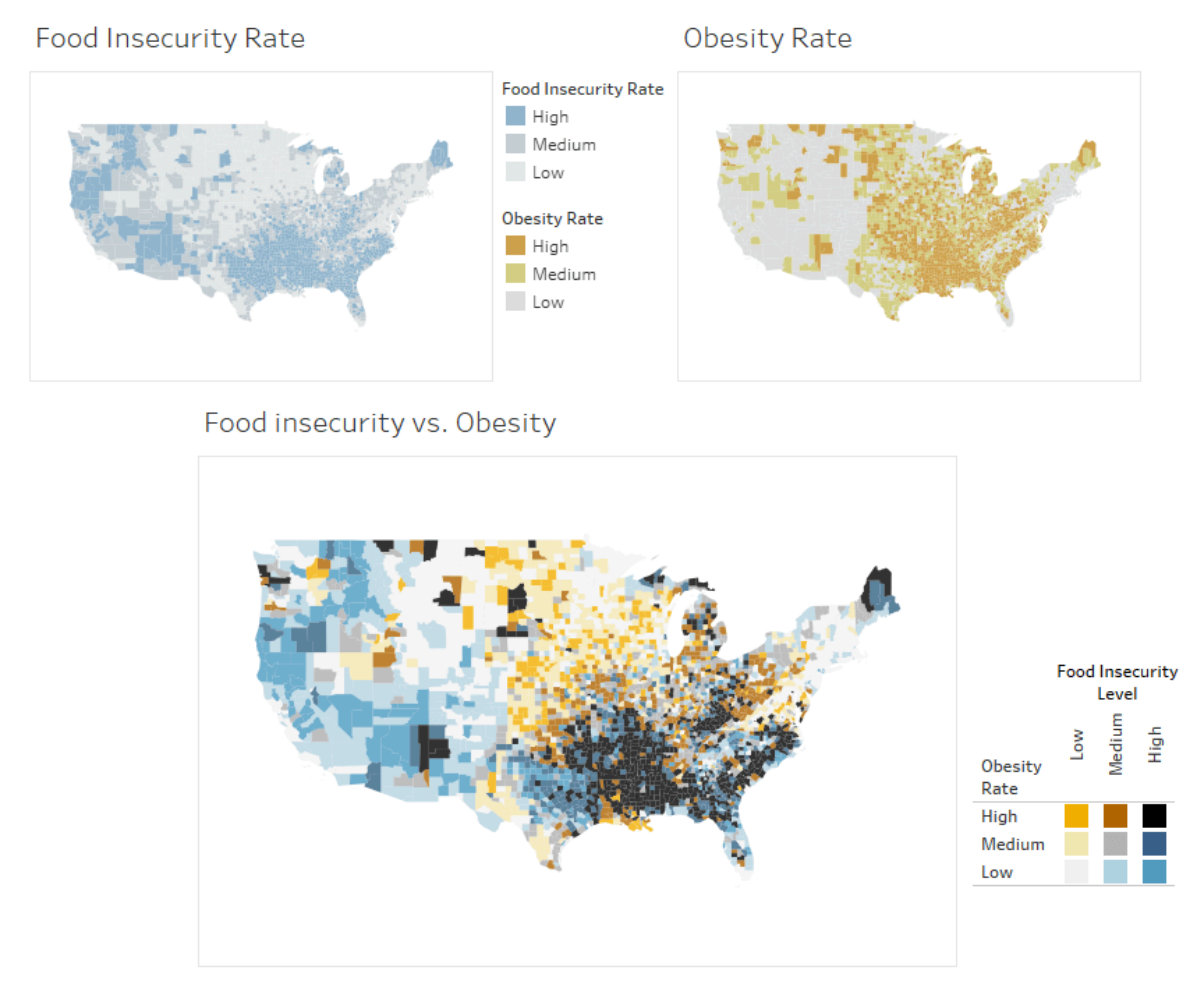
One categorical + one ordinal attribute



Crop type (categorical) is encoded by hue; **acreage class (ordinal)** by lightness.

Sarah Battersby · Tableau · Bivariate choropleth maps

Two ordinal attributes



This colormap encodes **two ordinal attributes**: food insecurity level and obesity level.

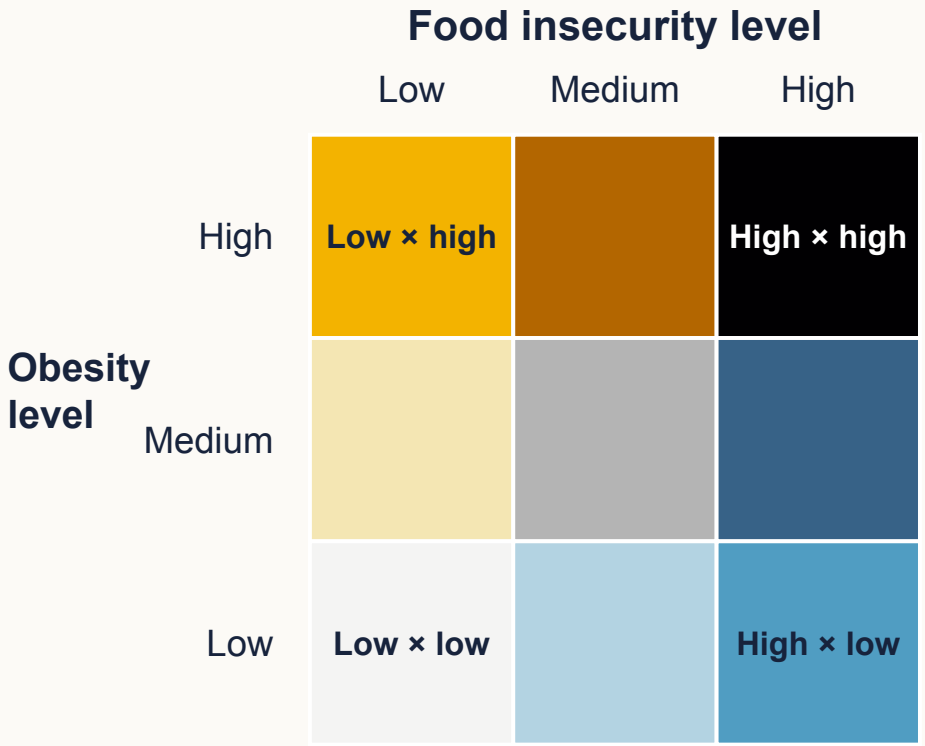
Each has **three ordered classes**, forming a 3 × 3 legend.

Read one axis for food insecurity and the other for obesity.

A cell represents a pair of classes, such as high insecurity and low obesity.

Sarah Battersby · Tableau · Bivariate choropleth maps

Read a bivariate legend by rows and columns



Pairs: food insecurity × obesity

Columns: food insecurity increases from left to right.

Rows: obesity increases from bottom to top.

Each cell represents a pair of ordered classes.

Practice: find the four corner combinations in the preceding map.

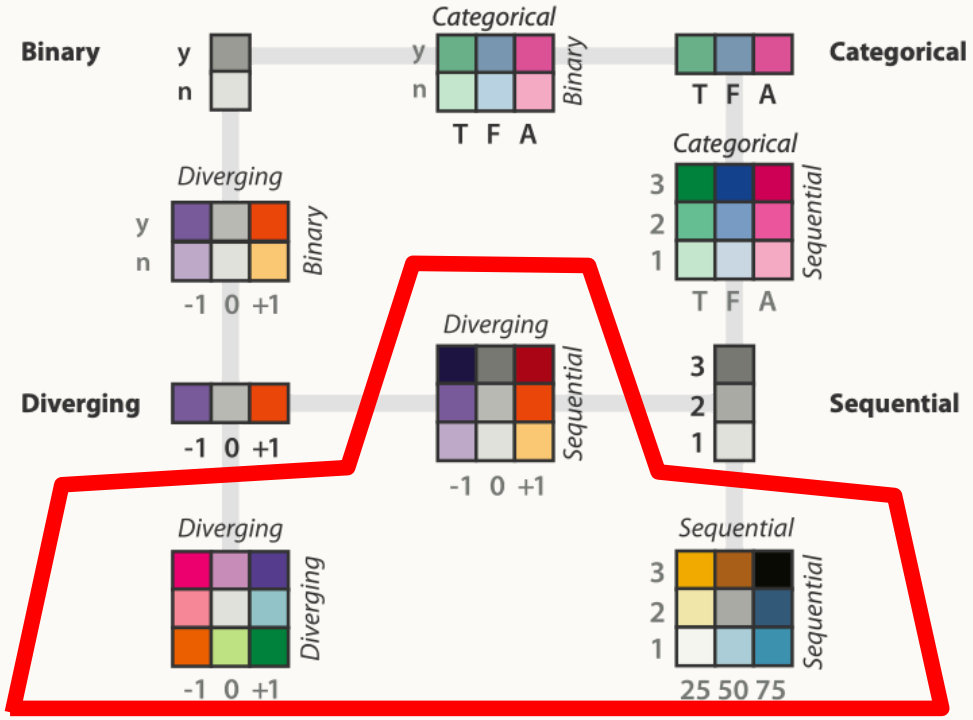
Lightness decreases from low–low to high–high, reinforcing the progression toward higher levels of both attributes.

Sarah Battersby · Tableau · Bivariate choropleth maps · Enlarged legend

Cases where you may use a bivariate colormap

- **Best:** one categorical attribute with two levels (binary) + one ordered attribute (sequential or diverging).
- **Middle:** two ordinal attributes (sequential or diverging).
- **Poorest:** two categorical attributes with multiple levels.

Always use with care.

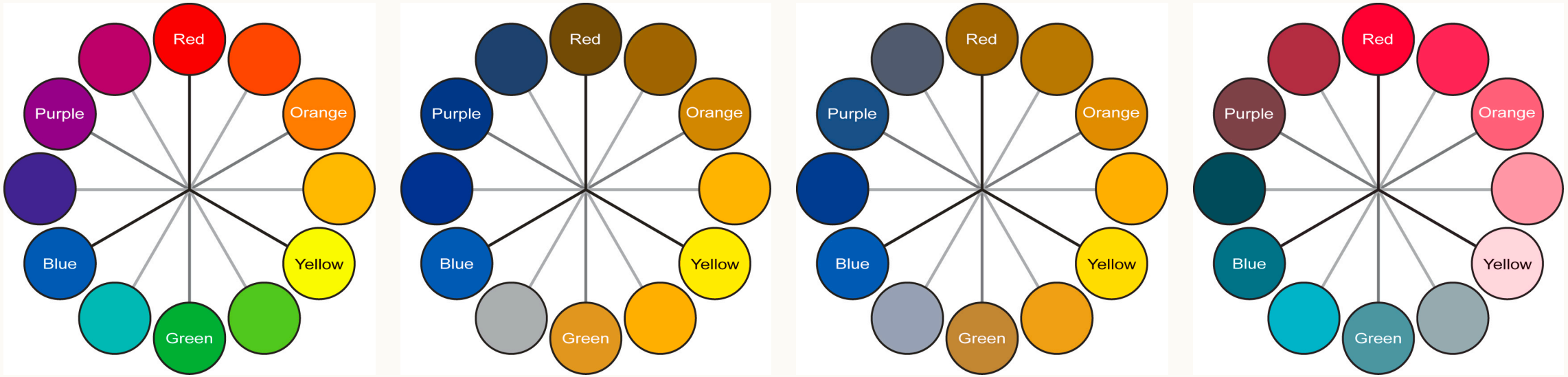


Use the combinations in the red frame with care: their meanings may be difficult to interpret.

Reading: Tamara Munzner · Color encoding · Tableau · Bivariate maps

Palette choice can mitigate color vision barriers

People with **deuteranopia** or **protanopia** can generally distinguish **blue from orange**, making blue–orange a useful alternative to red–green.

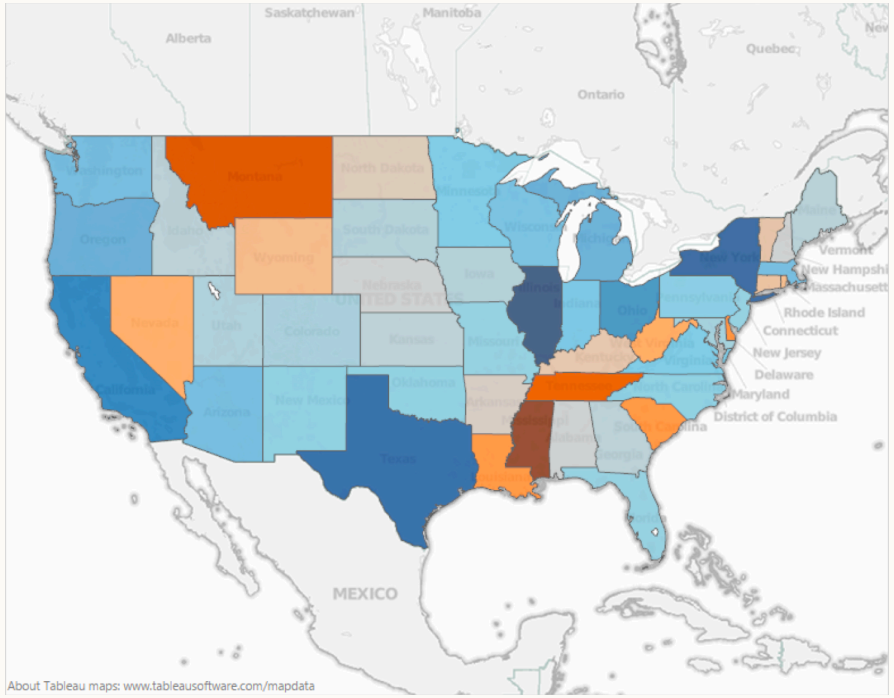


Original | Deuteranopia | Protanopia | Tritanopia

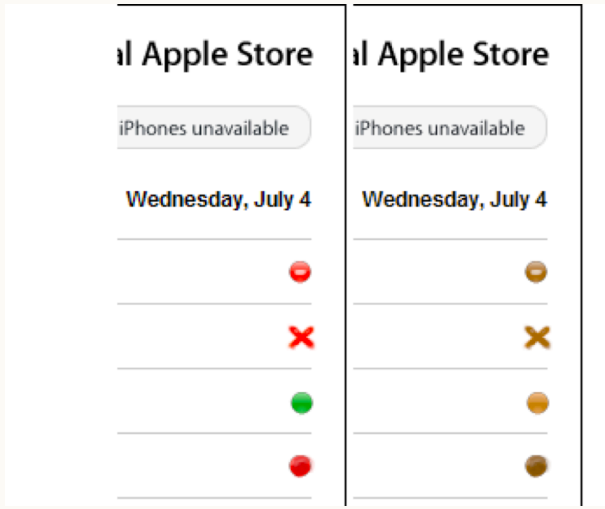
Use simulations to check the chosen shades, including for other deficiencies such as tritanopia.

Okabe & Ito · Color Universal Design · National Eye Institute · Types of Color Vision Deficiency

Color differences work better with another cue



Palette: blue–orange can avoid some common red–green confusions.



Redundancy: shapes and lightness preserve distinctions when hues become similar.

Keep important categories identifiable through labels, shapes, or patterns as well as color.

Jeffrey Shaffer · Tableau · Colorblind-friendly visualizations · W3C · Use of Color

Summary

1 **Color perception depends on context.** Background, spatial separation, culture, and color vision affect how colors are read.

2 **Match the palette to the data.** Hue distinguishes categories. Ordered palettes need a consistent lightness progression; diverging palettes need a meaningful reference.

3 **Define the numeric mapping.** Choose the domain, normalization, and class boundaries. Make their meaning explicit in the legend.

4 **Choose how many attributes to encode.** Bivariate colors represent pairs of values and need a readable two-dimensional legend.

5 **Check distinguishability and accessibility.** Test the colors in their intended context and reinforce important distinctions with labels, shapes, or patterns.

Appendix

Supplementary resources and examples for bivariate colormaps.

Two resources explain bivariate map construction

Tableau tutorial

How to make effective bivariate choropleth maps with Tableau

Examples, class combinations, and the cost of treating nine pairs as unrelated categories.

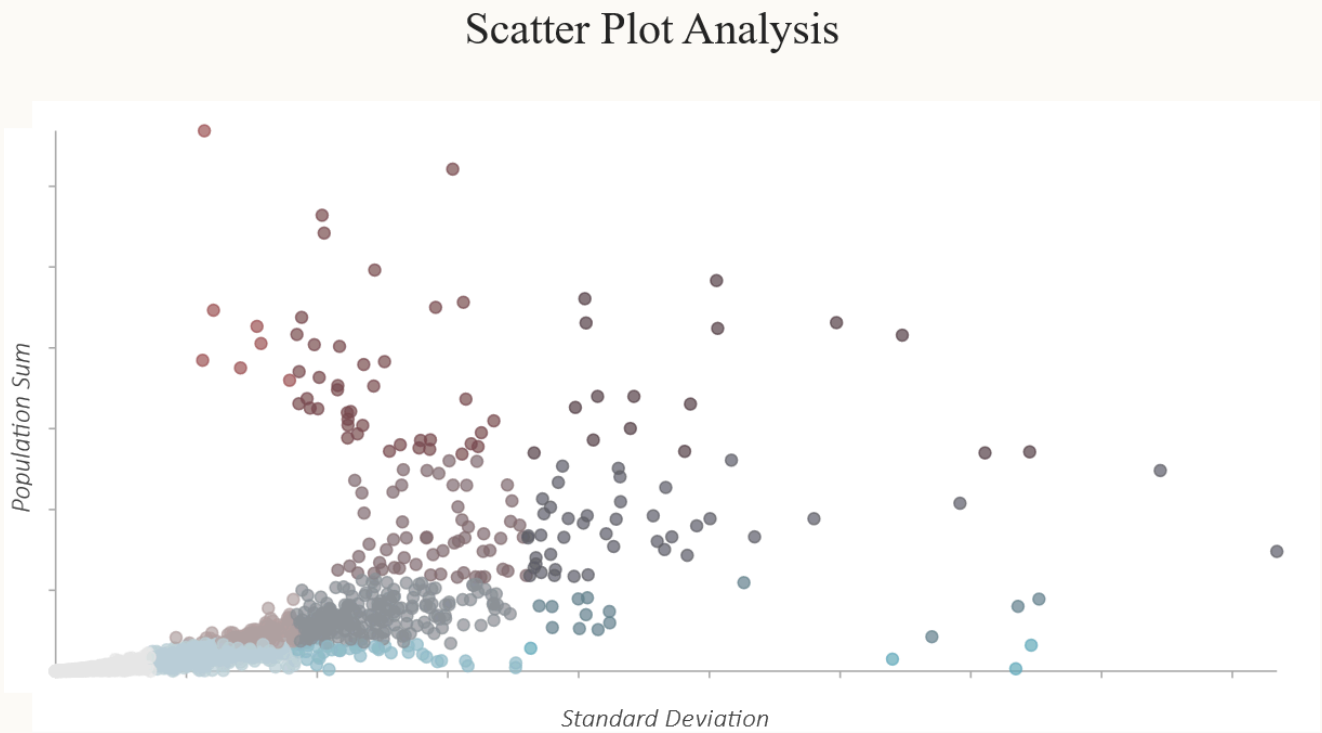
Observable notebook

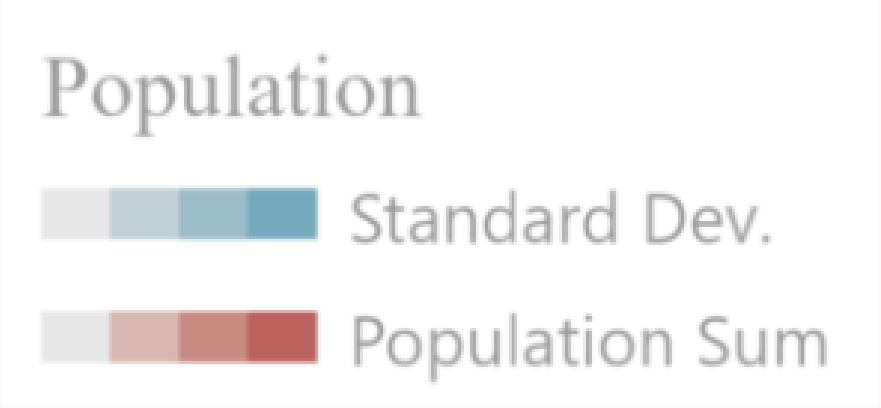
Bivariate Choropleth Mapping · awsherman

A worked reference for constructing a two-variable choropleth and its legend.

Sarah Battersby · Tableau (2018) · awsherman · Observable

Blending two colors can make both variables hard to read





The separate color ramps do not fully explain the **mixed colors** in the marks.

Use a joint legend, separate views, or another channel when the audience must recover both values.